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Topical Review

Assembly and integration of micro-led displays: a review of transfer methods targeting near-perfect yield

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Abstract

Micro-light-emitting diodes (micro-LEDs) are widely recognized for their superior brightness, efficiency, and durability, offering transformative potential for next-generation displays and emerging applications. However, the path to commercialization remains hindered by a critical barrier: achieving an extremely high production yield. Unlike conventional displays, micro-LED displays require the precise transfer of millions of individual micro-LED chips, even the slightest defects can significantly affect the overall yield. This review focuses on the technological challenges of pushing micro-LED assembly yield to extremely high thresholds of 99.999 99%. We explore six key transfer methods—elastomeric transfer, roll-to-roll printing, electrostatic and electromagnetic assembly, laser transfer, microvacuum assembly, and fluidic self-assembly—and analyze their respective impacts on yield. Recent advancements in each method are discussed, with an emphasis on strategies to overcome yield challenges. By framing yield as the central metric and not as a side concern, this review aims to provide a roadmap for overcoming bottlenecks in micro-LED assembly and enabling industrial-scale deployment.

Keywords: assembly, integration, micro-led displays, transfer method, yield

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1. Introduction

Micro-LEDs are poised to revolutionize display and lighting technologies because of their superior brightness, efficiency, and longevity compared to conventional display technologies such as liquid-crystal displays (LCDs) and organic light-emitting diodes (OLEDs) (Table 1)^[1–14]. Micro-LEDs offer several key advantages, including higher brightness, lower energy consumption, better color fidelity, and faster response times. These characteristics make micro-LEDs highly attractive for a range of applications, including next-generation displays, wearable devices, automotive displays, and biomedical devices^[15–31]. As the demand for higher-performance displays increases, micro-LED technology has advanced rapidly, driven by its potential for mass adoption in various industries. However, despite their promising features, the commercialization of micro-LEDs remains a significant challenge.

One of the most significant challenges in micro-LED technology is the transfer yield of the assembly process, which requires precise transfer and integration of millions of micro-scale LED chips onto a display substrate. These micro-LED chips, typically smaller than 100 μm , must be accurately placed and aligned on the substrate, often with pixel pitches below 10 μm . Achieving this level of precision at high speeds, particularly on the scale necessary for mass production, is extremely challenging. Furthermore, the demand for high throughput, which is the ability to process large volumes of chips efficiently, combined with the need for high yield, which involves minimizing defects, further complicates assembly^[32–43]. This has increased the costs and slowed down the commercialization of micro-LED technology. Various techniques for micro-LED assembly have been developed, each offering advantages and drawbacks. The six major methods are elastomeric transfer printing, roll-to-roll printing, electrostatic assembly, electromagnetic assembly, laser-based transfer, and fluidic self-assembly. These methods address the key challenges of precision, scalability, cost efficiency, and transfer yield in micro-LED production to improve the assembly process for large-scale applications.

These methods differ significantly in terms of cost. Elastomeric stamp transfer offers low equipment costs but incurs moderate consumable expenses due to frequent stamp replacements. Roll-to-roll methods, while ideal for large-scale production, demand a high initial investment in specialized machinery. Electrostatic and electromagnetic transfers involve relatively high equipment costs and require precise control, which increases overall process complexity despite minimal consumable use. Laser-assisted transfer is among the most expensive due to the need for high-power lasers and optical systems, but it delivers excellent selectivity and precision for fine-pitch applications. Microvacuum transfer provides a cost-effective alternative for high-precision applications, combining low overall cost and low chip damage. Finally, fluidic self-assembly is inexpensive in terms of equipment but can incur additional costs for fluidic media optimization. Ultimately,

the cost-effectiveness of each method depends on production scale, resolution requirements, and yield targets.

Light-emitting diodes (LEDs) are semiconductor devices that emit light when bias voltage is applied^[44–48]. An LED is essentially a p-n diode with a heterojunction structure. When a forward voltage is applied, recombination occurs as the holes and electrons meet, releasing energy as photons. LEDs are single-color light sources that emit light at specific wavelengths. The color of the emitted light is determined by the bandgap energy, which is the difference between the conduction and valence bands. A larger bandgap produces shorter wavelength light, whereas a smaller bandgap produces longer wavelength light. The bandgap energy and light color depend on the materials used, such as GaN, GaAs, and GaP. AlGaInP is used for red-to-yellow light, whereas InGaIn is used for green-to-blue light. Both types are widely used in industry for various applications.

Micro-LEDs were initially defined as inorganic light-emitting diodes (iLEDs) with chip sizes under 100 μm ^[49–62]. Because mini-LEDs now appear at similar sizes, distinguishing them is necessary to define micro-LEDs better. The key difference lies in the display roles. In some studies, mini-LEDs are used as pixel units for stretchable displays, whereas in LCDs, they mainly serve as backlights, improving contrast and power efficiency through local dimming, but do not act as pixels^[63–69]. In contrast, micro-LEDs emit light directly and serve as pixels, making them crucial for high-resolution displays such as augmented reality (AR)/mixed reality (MR) needing over 3 000 pixels per inch (PPI)^[70–74]. To achieve this, micro-LEDs must shrink to 10 μm and become thinner, requiring separation from the sapphire substrate^[75–86]. As a result, micro-LEDs are now defined as substrate-free iLEDs with chip sizes of 75 μm or smaller^[87,88].

Micro-LED technology has been adopted in various display applications including signage^[89,90], televisions (TVs)^[91–94], monitors^[95–98], automotive displays^[99–101], smart watches^[102–105], and AR glasses^[106–114]. Although these applications share the same core technology, the assembly process differs depending on how the micro-LEDs are arranged as pixels and integrated into a display. These differences arise from variations in the pixel configuration, substrate materials, and manufacturing constraints. Understanding these variations in assembly requires a clear understanding of their fundamental characteristics (Table 2). One of the parameters in display performance is the viewing distance (D), which refers to the optimal distance between the viewer and screen for a clear visual experience^[115–120]. It depends on the display size (S), measured as the diagonal length of the screen, as well as the viewing angle (θ) and the aspect ratio ($R_W:R_H$), which defines the proportional relationship between width and height:

$$\text{Viewing Distance : } D = \frac{0.5 \times W}{\tan\left(\frac{\theta}{2}\right)} \quad (1)$$

Table 1. Comparison of LCD, OLED, and micro-LED.

Parameter	LCD	OLED	Micro-LED
Mechanism	Backlight	Self-emissive	Self-emissive
Pixel density	1 000 PPI	2 500 PPI	5 000 PPI
Luminous efficiency	Medium	Low	High
Luminance ($\text{cd}\cdot\text{m}^{-2}$)	3 000 (full color)– 10^4 (green)	1 500 (full color)– 10^3 (yellow)	10^5 (full color)– 10^7 (blue/green)
Contrast ratio	Medium (5 K:1)	High (10 K:1)	Very high (1 M:1)
Response time	Low (ms)	High (μs)	Very high (ns)
Operating temperature	0 °C–60 °C	–50 °C–70 °C	–100 °C–120 °C
Viewing angle	Low	High	High
Shock resistance	Low	Medium	High
Environmental stability	Good	Medium	Best
Flexibility	Low	High	Medium
Lifetime	Good	Medium	Best
Energy consumption	High	60%–85% of LCD	30%–40% of LCD
Maturity	High	Medium	Low
Cost	Low	Medium	High

Table 2. Comparison of display characteristics for different micro-LED applications.

Application	Viewing distance	Display size /inch	Chip type	Colour	Chip size/ (L × W, μm)	Resolution (Pixel _{width} × Pixel _{Height})	Pixel pitch/ μm	PPI (Pixels per inch)	References
Signage	4.39 m	144	Flip-chip	RGB	182 × 93	FHD (1 920 × 1 080)	840	30	[89,90]
TV	3.47 m	114	Flip-chip	RGB	34 × 85	4 K+ (4 968 × 2 808)	500	50	[91–94]
Monitor	60 cm	17.3	Flip-chip	RGB	<100	720p (1 280 × 720)	300	85	[95–98]
Automotive	75 cm	13.5	Flip-chip	RGB	<30	FHD (1 920 × 1 080)	156	163	[99–101]
Smart watch	35 cm	2.1	Flip-chip	RGB	<30	418 × 540	78	326	[102–105]
AR glass	0.5 cm	0.13	Monolithic	Monochrome	4 × 4	640 × 480	4	6 350	[106–113]

The width (W) of the display can be derived from the diagonal size using trigonometric relationships as follows:

$$\text{Width of display} : W = S \times \frac{R_W}{\sqrt{R_W^2 + R_H^2}}. \quad (2)$$

Another fundamental parameter is resolution, which quantifies the total number of pixels that composes an image and is typically expressed as pixel width (W_p) × pixel height (H_p) (e.g., 1 920 × 1 080 for Full HD). Although resolution indicates the pixel count in a display, it does not directly define pixel density, which is a key factor in determining image sharpness and clarity. Instead, the pixel density is better expressed through PPI and pixel pitch. PPI represents the number of pixels present in a one-inch length of the display, whereas pixel pitch defines the physical spacing between adjacent pixels.

$$\text{PPI} = \frac{\sqrt{W_p^2 + H_p^2}}{S}. \quad (3)$$

Because the PPI and pixel pitch are inversely related, a smaller pixel pitch corresponds to a higher pixel density. A pixel is the smallest addressable unit in a display and represents a single point in an image. However, each pixel is composed of subpixels emitting red, green, and blue light,

which combine to produce the desired color. Their arrangement affects the color mixing. Pixel pitch limits the maximum pixel size, thereby constraining the subpixel dimensions. For example, if the pixel pitch is 300 μm , the pixel size must be 300 μm or smaller, and each subpixel must be less than 100 μm . These structural relationships influence the display design and integration strategies.

Micro-LED integration methods vary depending on the pixel density required for the target application^[121]. In low-to-mid PPI displays, where pixel pitch is relatively large, horizontally arranged RGB subpixels can be transferred onto the display driver integrated circuit (DDIC) using transfer methods (Figure 1(a))^[122–126]. This approach allows each micro-LED to be picked up and transferred individually or in groups, then bonded using a conductive adhesive layer to ensure electrical connection to DDIC providing electrical and mechanical stability. In addition, this integration approach offers the potential for heterogeneous integration with devices produced in varied manufacturing environments, thereby broadening the design flexibility and functionality of the resulting display systems^[127–136].

In contrast, high-PPI applications like AR displays require much tighter subpixel spacing and often favor vertical stacking of RGB subpixels to maximize integration density (Figure 1(b)). Here, wafer bonding-based hybrid monolithic integration enables precise alignment and vertical pixel

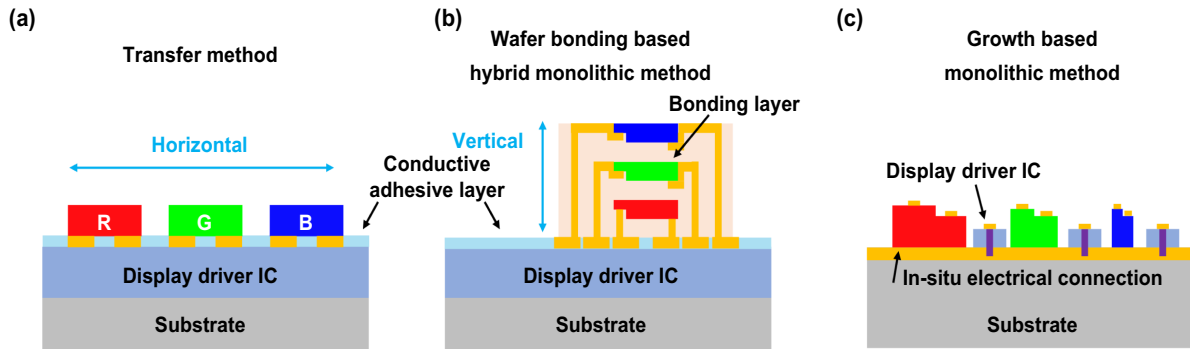


Figure 1. Micro-LED pixel arrangement and integration with DDIC using various methods. Reproduced from^[121]. CC BY 4.0. (a) Transfer method. (b) Wafer bonding-based hybrid monolithic integration. (c) Growth-based monolithic integration.

stacking, allowing for more compact layouts beyond the resolution limits of traditional transfer methods^[137–146].

Finally, some emerging approaches aim to monolithically grow both the micro-LEDs and DDICs on a single wafer (Figure 1(c)). In this growth-based method, RGB micro-LEDs and transistors are fabricated through sequential epitaxial growth and in-situ processing to retain electrical connectivity within the same substrate^[147–158]. Although full color realization through this method remains technically challenging, it represents a promising route toward highly integrated and compact display systems.

Despite these advancements, AR glasses still face scalability challenges. Conversely, lower- to mid-PPI applications, such as signage and TVs, might seem less challenging, but present unique assembly hurdles. These applications require the transfer of densely packed micro-LEDs from wafers to larger-pixel-pitch panels. Although individual LEDs are larger, achieving a high-yield defect-free transfer remains a challenge owing to alignment constraints and substrate variations. To address this, transfer techniques such as elastomeric stamping and laser-assisted methods are being developed to improve efficiency and scalability. For high-PPI microdisplays or large-area panels, micro-LED assembly requires specialized techniques tailored to manufacturing constraints, driving advancements in display technology.

Key milestones in micro-LED transfer methods have played a crucial role in the evolution of these methods over the past two decades (Figure 2). Elastomeric stamp transfer^[124,159,164–205] utilizes an elastomeric stamp to pick-up and place ink on a target substrate through van der Waals (vdW) interactions (Figure 2(a)). Initially, this technique was demonstrated for transferring silicon microstructures and solar cells, but over time, it has increasingly been applied to micro-LED transfer technologies as well^[196]. This method is efficient and economical, allowing high-throughput transfer while maintaining low production costs. Additionally, it requires no voltage or heat, making it compatible with thermally sensitive materials and minimizing the potential damage to micro-LEDs. However, poor repeatability remains a major drawback, as inconsistencies in the adhesion strength lead to variations in the transfer yield. Another significant challenge is residue

and contamination, where micro-LEDs may not detach cleanly from the stamp, leaving behind unwanted materials that can degrade the display performance. Despite these drawbacks, elastomeric stamp transfer has served as a foundational technique providing insights into the development of advanced micro-LED transfer methods.

Roll-to-roll transfer^[160,206–213] is a micro-LED assembly technique in which micro-LEDs are continuously transferred from a donor substrate to a receiver substrate via a rolling process (Figure 2(b)). A key advantage of this method is its cost-effectiveness, as it enables mass production and significantly reduces unit manufacturing costs. Additionally, its high throughput and efficiency allow the rapid processing of micro-LEDs, making them suitable for large-scale applications. However, mechanical stress can damage the micro-LEDs because the rolling mechanism introduces deformation and potential defects. Another challenge is the alignment precision, which requires sophisticated control systems to ensure accurate micro-LED placement. Despite these challenges, roll-to-roll processing remains a promising solution for scaling up micro-LED manufacturing, offering a balance between throughput, cost, and efficiency.

Electrostatic^[214–220] and electromagnetic^[221] transfer utilizes electric or magnetic forces to selectively transfer micro-LEDs, offering enhanced precision and non-contact assembly (Figures 2(c) and (d)). A key advantage of this method is its flexibility and perfect repeatability, as the electrostatic and electromagnetic forces can be precisely tuned for consistent and controlled transfer. Additionally, it eliminates residue and contamination, as it avoids mechanical contact, which can introduce defects. However, a high voltage can lead to LED breakdown, limiting material compatibility, and increasing process complexity. Another drawback is its small transfer area, which makes it less effective for large-scale applications requiring high-throughput processing. Nonetheless, these transfer methods provide a non-contact alternative, reducing the risk of mechanical stress and enabling greater precision in micro-LED assemblies.

Laser^[161,222–235] transfer printing utilizes laser pulses to release and place micro-LEDs on a substrate with high accuracy (Figure 2(e)). One of its key strengths is that

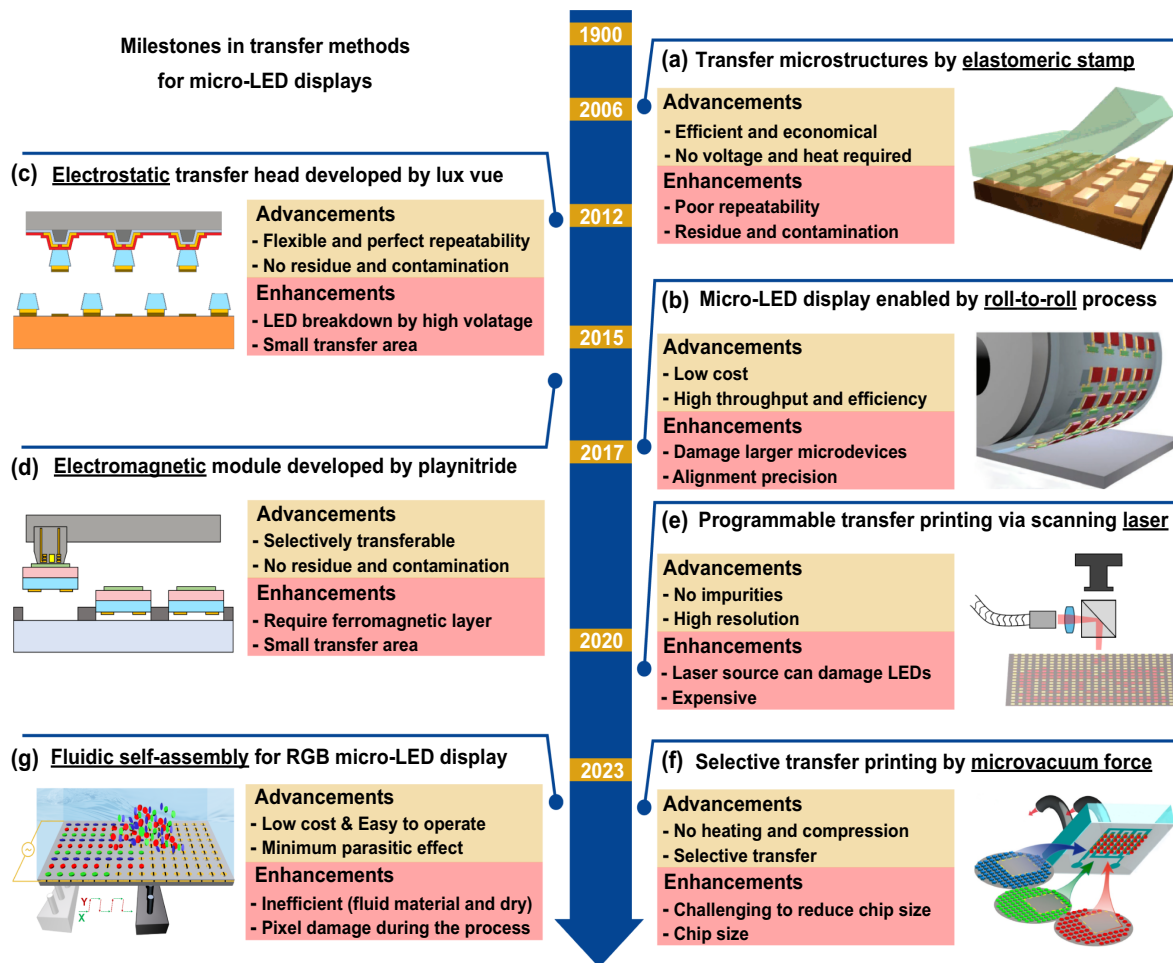


Figure 2. Major milestones in transfer methods for micro-LED displays and their advantages & enhancements. (a) Elastomeric stamp. Reprinted (adapted) with permission from^[159]. Copyright (2007) American Chemical Society. (b) Roll-to-roll. ^[160] John Wiley & Sons. © 2017 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (c) Electrostatic. (d) Electromagnetic. (e) Laser. Reproduced with permission from^[161]. Copyright © 2020 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC). (f) Vacuum. Reproduced from^[162]. CC BY 4.0. (g) Fluidic self-assembly-based techniques. Adapted from^[163], with permission from Springer Nature.

it introduces no impurities because the laser-based process avoids direct mechanical contact and prevents contamination during transfer. Moreover, its high resolution allows precise micro-LED placement, supporting ultrafine pixel pitches essential for high-definition displays. However, laser exposure can damage LEDs, especially for delicate or small-scale devices, which require optimization to minimize thermal effects and material degradation. Additionally, the process is expensive because high-power lasers and precise optical setups significantly increase manufacturing costs. Despite these challenges, laser transfer printing remains a critical technique for advanced micro-LED assembly, particularly in high-resolution applications, where precise patterning and minimal contamination are crucial.

Microvacuum^[162,236] force transfer employs a localized vacuum pressure to selectively pick-up and transfer micro-LEDs with controlled precision (Figure 2(f)). A major advantage is that no heating or mechanical compression is required to prevent thermal or mechanical stress that can damage delicate

devices. In addition, its selective transfer capability enables the precise placement of individual micro-LEDs, making it particularly well-suited for assembling complex display architectures. However, reducing the chip size remains a challenge as vacuum forces become less effective at microscale dimensions, limiting scalability. Another concern is the overall chip size, which influences the vacuum efficiency and transfer precision, and requires careful optimization for consistent results. Despite these limitations, microvacuum force transfer remains a promising approach for achieving high-precision micro-LED assemblies, particularly for applications requiring minimal mechanical stress and high placement accuracy.

Fluidic self-assembly^[163,237–240] relies on capillary forces to guide micro-LEDs to predefined positions, offering a scalable approach for large-area display integration (Figure 2(g)). A major benefit is its low cost and ease of operation because the self-alignment process minimizes the need for complex equipment, reduces production costs, and simplifies large-scale implementation. Additionally, the minimal parasitic

effect ensures efficient material utilization and precise positioning, thereby optimizing the display uniformity and yield. However, inefficient material handling, both in wet and dry states, can reduce the overall transfer efficiency, requiring further refinement to enhance process control. Another drawback is the risk of pixel damage because uncontrolled fluid dynamics can introduce mechanical stress and misalignment. Despite these issues, fluidic self-assembly presents a novel route for integrating micro-LEDs into next-generation displays, balancing cost-effectiveness with scalability while improving manufacturing efficiency.

2. Elastomeric stamp-based techniques

Elastomeric stamp-based techniques represent one of the earliest and most extensively studied micro-LED transfer methods, and they continue to attract significant research interest today^[164–168,201–205]. The elastomeric stamp is a crucial tool in a micro-LED assembly that utilizes polydimethylsiloxane (PDMS), a unique and versatile viscoelastic material, for precise device transfer. PDMS, which is formed by combining a base and a curing agent, hardens gradually while maintaining its elasticity. This material is valued for its ability to conform to complex surfaces and withstand repeated usage. Its slightly tacky surface, created by the balance between elasticity and mild adhesion, allows it to pick-up and transfer micro-LEDs without additional adhesives. This adhesion, primarily due to vdW forces, ensures efficient handling with minimal mechanical stress, because the softness of the material reduces the risk of damaging delicate components during transfer.

The versatility of the stamp allows customization into various shapes, enhancing adaptability across a wide range of applications^[169–187]. Furthermore, elastomeric stamps are compatible with other advanced assembly techniques, including laser-, roll-to-roll-, and vacuum-based methods, thereby broadening their potential and supporting high-throughput manufacturing in optoelectronics and flexible electronics. The cost-effective, easy-to-produce, and scalable PDMS-based stamps provide a reliable solution for micro-LED assembly by combining efficiency, adaptability, and compatibility with other transfer technologies. This section introduces methods to control the adhesion between micro-LEDs and stamps during pick-up and printing by optimizing the stamp structures and transfer motions.

2.1. Kinetic control of adhesion to elastomeric stamp

In 2006, Rogers *et al* developed an assembly technology using an elastomeric stamp made of PDMS. This technique achieves precise kinetic control of adhesion to an elastomeric stamp by manipulating the peeling speed during the assembly process^[159,188,189]. The process starts with the contact between the stamp and ink on the donor substrate (Figure 3(a)). When the stamp was peeled away at a high speed, the inks were picked up from the donor substrate to the

stamp. The ink was then aligned and placed in contact with the receiver substrate. Finally, as the stamp was peeled away at low speed, the inks were printed onto the receiver substrate. The fundamental mechanism relies on the vdW forces and viscoelastic behavior of the elastomeric stamp, making the adhesion between the stamp and ink speed-dependent. This adhesion variation acts as a switch to determine whether the ink is picked up or released, thereby enabling reliable transfer. When the PDMS stamp is peeled from the donor substrate at high velocities, typically approximately $10 \text{ cm}\cdot\text{s}^{-1}$ or faster, the adhesion becomes strong enough to pick-up the ink securely. At lower velocities, such as $1 \text{ mm}\cdot\text{s}^{-1}$ or lower, the adhesion is weakened, allowing the ink to be released onto the receiver substrate. To explain this mechanism, the adhesion energy G follows the following relationship:

$$G(v) = G_0 \left[1 + \left(\frac{v}{v_0} \right)^n \right] \quad (4)$$

where G_0 is the adhesion energy at zero peel velocity, v is the peel velocity, v_0 is the reference velocity, and the component n is a scaling parameter determined experimentally. This formula shows how the adhesion energy increases with the peel velocity. Based on this theory, the critical velocity v_c defines the transition between pick-up and printing during elastomeric transfer printing. The graph illustrates the relationship between the peel velocity v (x -axis) and the adhesion energy G (y -axis) (Figure 3(b)). The red line in the middle of the graph serves as a reference for the adhesion energy G_c of the ink/substrate interface, whereas the curve represents how the adhesion energy for the ink/stamp interface varies with velocity. At v_c , G surpasses G_c , which determines whether the stamp picks up the ink or releases it onto the receiver substrate. This speed-dependent mechanism enables a precise and controllable ink transfer. The conditions for pick-up and printing are determined by comparing the adhesion energy. Pick-up occurs when:

$$G_c < G(v) \quad \text{for pick - up} \quad (5)$$

and printing occurs when:

$$G_c > G(v) \quad \text{for printing.} \quad (6)$$

The kinetically controlled elastomeric stamping technique has evolved through further research, leading to the development of microstructured PDMS stamps with improved transfer efficiencies and reliabilities (Figure 3(c))^[127,190–196]. These stamps feature an array of microscale protrusions that deform upon contact, thereby increasing the effective contact area and adhesion during pick-up. Upon retraction, the protrusions recover their original shape, significantly reducing the adhesion and enabling precise material release onto the target substrate. Unlike conventional flat PDMS stamps, which provide uniform adhesion across their surfaces, microstructured stamps introduce an additional level of tunability by

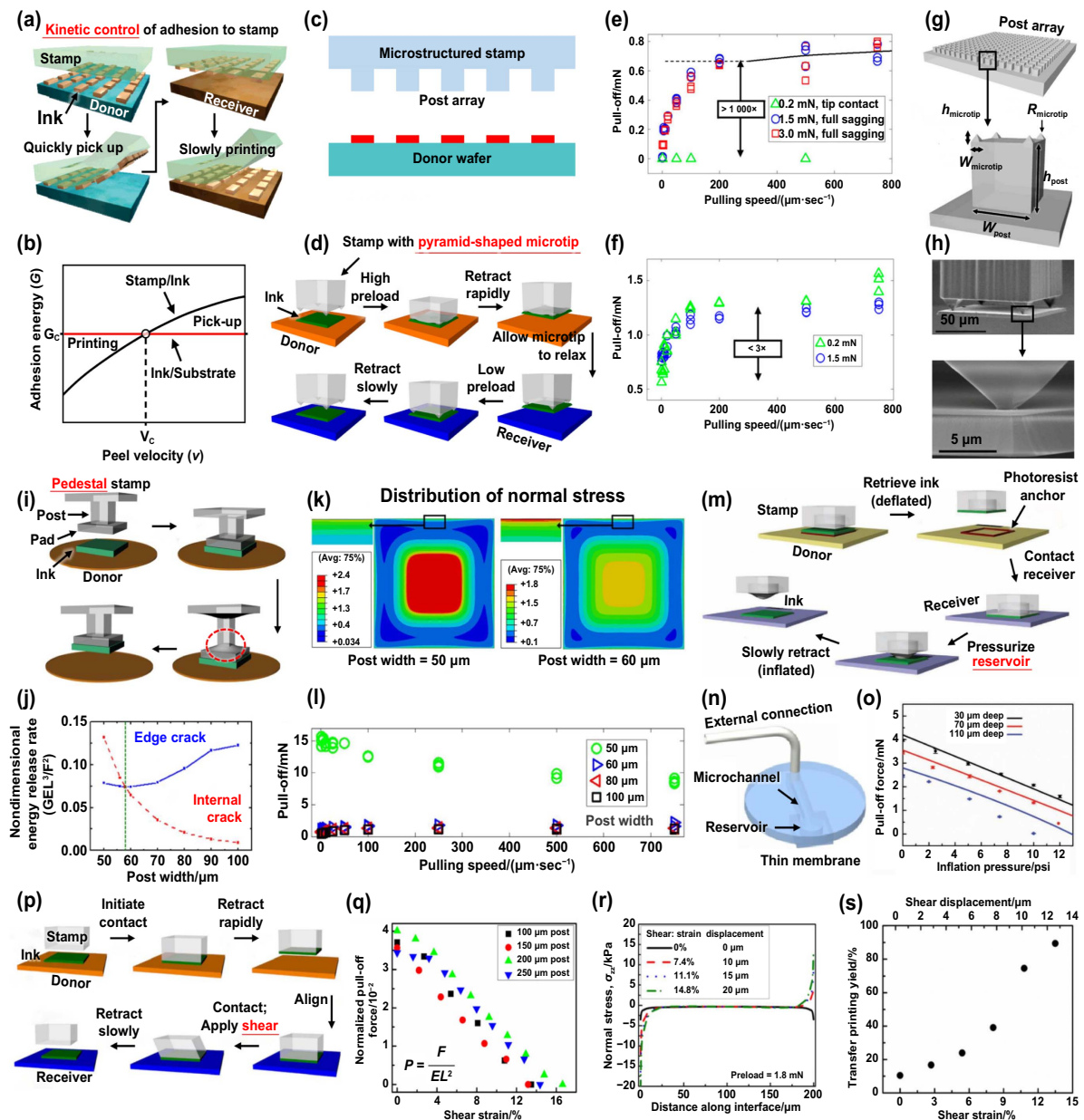


Figure 3. Elastomeric stamp-based techniques. (a) Transfer printing process utilizing kinetic control. Reprinted (adapted) with permission from [159]. Copyright (2007) American Chemical Society. (b) Adhesion comparison based on peel velocity to distinguish between pick-up and printing stages. (c) Microstructured stamp composed of square post arrays. (d) Transfer printing process using an improved microstructured stamp with a pyramid-shaped microtip. (e) Pull-off force graph comparing tip contact and full sagging conditions at varying pulling speeds. (f) Pull-off force graph for a flat stamp under different pulling speeds. (g) Specifications of the stamp and microtip dimensions. (h) SEM image of a stamp in micro-tip design with silicon platelet. Reproduced with permission from [197]. (i) Pick-up process using a pedestal stamp. (j) Comparison of edge and internal delamination based on post width. (k) Stress distribution on the stamp pad depending on post width. (l) Pull-off force graph comparing different post widths at various pulling speeds. Reprinted from [198], with the permission of AIP Publishing. (m) Transfer printing process with a reservoir-embedded stamp. (n) Pathway of pressure fluid movement across the entire stamp structure. (o) Pull-off force graph by pressure with varying reservoir depths. [199] John Wiley & Sons. Copyright © 2012 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim (p) Shear-force-enhanced transfer process. (q) Normalized pull-off force graph by shear strain, independent of post width. (r) Stress distribution on the stamp interface under different shear strain levels. (s) Yield graph based on shear strain. Reprinted from [200], with the permission of AIP Publishing.

leveraging both preload and peel velocity controls, allowing for more precise manipulation of adhesion forces. This advancement improves the scalability, enabling the parallel transfer of large device arrays while maintaining high throughput, cost efficiency, and transfer accuracy, even for complex

patterns. Additionally, with transfer yields exceeding 99.5%, this method has proven to be effective in applications such as micro-LED assembly, flexible electronics, and optoelectronic devices, where precise material placement and alignment are critical for performance and reliability. This technique

demonstrates that even the peeling velocity of the stamp can critically modulate adhesion, highlighting motion control as a key factor in transfer yield. We believe this provides an important perspective that the dynamic behavior of the transfer head—such as speed and acceleration—can influence yield optimization not only in elastomeric stamp methods but also in other transfer techniques.

2.2. Microstructured elastomeric stamp with microtips

In 2010, Rogers *et al* introduced an improved version of a microstructured elastomeric stamp^[197]. This stamp consists of a square array of square posts, each precisely designed with a pyramid-shaped microtip positioned at the corners of the top surface (Figure 3(d)). This process demonstrates how the microtips undergo mechanical collapse and recovery under different preload conditions during pick-up and printing. Preload refers to the force applied while pressing the stamp against the substrate before lifting, determining whether the ink is picked up or released. During pick-up, a high preload was applied to ensure strong adhesion between the stamp and ink. As the stamp approached the ink on the donor substrate, the microtips made initial contact, forming a small contact area that expanded as the preload increased. This leads to post-contact and eventual zipping of the interface, where the contact radius of the tip nearly matches the pyramid base and the post-surface fully engages with the ink. At this point, the rapid retraction generates a strong pull-off force that lifts the ink from the donor onto the stamp. After picking up, the compressed microtips gradually relaxed over time, reducing the contact area and leaving the ink attached only to their tips. The ink remains at the microtip ends with minimal force during the approach. Upon lifting the stamp, the weak adhesion at the tip interface allowed the ink to be released onto the receiver substrate. This preload-dependent mechanism creates a pull-off force difference of more than 1 000-fold between pick-up and printing.

The graphs show the pull-off force (y-axis) as a function of the pulling speed (x-axis) for the microtip and flat stamps under different preloading conditions (Figures 3(e) and (f)). During pick-up, both the microtip and flat stamps exhibited successful ink retrieval at high pulling speeds. The microtip stamp requires a high preload (1.5 mN, blue circles; 3.0 mN, red squares) to collapse the microtips and establish full contact with the ink, resulting in a pull-off force exceeding 0.6 mN at pulling speeds above $200 \mu\text{m}\cdot\text{s}^{-1}$. In the case of the flat stamp, a similar trend is observed across both low (0.2 mN, green triangles) and high (1.5 mN, blue circles) preload conditions, as the ink remains in full contact with the flat post surface, leading to a pull-off force exceeding 1.0 mN at high pulling speeds. During printing, the microtip stamp operates under a low preload (0.2 mN, green triangles), where only the microtip makes contact. Consequently, the pull-off force remains near zero. In the case of the flat stamp, a pull-off force exceeding 0.5 mN was recorded even at very slow pulling speeds under both low and high preload conditions. This comparison reveals that the microtip stamp offers exceptionally high levels

of switching in adhesion, with over a 1 000-fold difference in the pull-off force between pick-up and printing, compared with less than a 3-fold difference for the flat stamp. The stamp design was optimized and validated in three steps: (1) theoretical modeling and finite element modeling (FEM) simulations to determine the optimal values for the microtip height, tip contact radius, post width, and tip configuration (Figure 3(g)); (2) fabrication of the stamp with the optimized structure; and (3) validation through transfer experiments and SEM image analysis. The stamp was fabricated using a combined mold, where the silicon (Si) mold defined the microtips and the SU-8 mold defined the posts. The microtip height was determined by material properties and structural dimensions, with FEM optimizing the range to 8.44–13.3 μm . If below the $h_{\min}$ (8.44 μm), slow recovery after compression hindered adhesion switching, while exceeding the $h_{\max}$ (13.3 μm) led to excessive restoring force, destabilizing pick-up. The microtips were fabricated at 10.6 μm , adjusting the mask opening size during silicon patterning, where larger openings resulted in taller microtips. The tip contact radius was predicted to be 732 nm (FEM) and was confirmed to be 750 nm (SEM) (Figure 3(h)). The post width matched the $100 \mu\text{m} \times 100 \mu\text{m}$ ink size. The microtip configuration was adjusted based on ink thickness, with thin ink (260 nm) requiring a 5-tip design and thicker ink (3 μm) using a 4-tip layout. This optimized design enabled reliable transfer printing with a very high yield (almost 100%). This study is significant in that it focuses on minimizing printing adhesion, one of the critical factors affecting transfer yield. By effectively weakening the adhesion during the printing step, it contributes to more reliable and high-yield transfer processes.

2.3. Pedestal-shaped elastomeric stamp

In 2012, Rogers *et al* developed a pedestal-shaped elastomeric stamp that significantly enhanced the pick-up process in transfer printing by providing higher adhesive forces^[198]. The pedestal structure consisted of a pad connected to a vertical post, forming a unique mechanical design. As depicted in the schematic of the pick-up procedure, when the pedestal stamp is lifted, the center of the pad is raised owing to its connection to the post (Figure 3(i)). This causes a concentration of adhesive forces in the central region of the pad, allowing for stronger attachment compared to traditional flat stamps, which typically exhibit weaker forces owing to edge-initiated delamination. The pedestal-shaped stamp is fabricated by separately molding the pad and post using SU-8 epoxy, followed by PDMS casting and partial curing to maintain the structural integrity. The two components are then precisely aligned and bonded for the final curing. The enhanced adhesion mechanism of the pedestal stamp was due to the formation and propagation of cracks during the detachment of the materials. The term “crack” refers to separation at the interface between the stamp and the material being transferred, rather than physical damage to the stamp itself. In conventional flat stamps, detachment begins at the outer edges of the pad, where cracks form and propagate inward, causing the contact area to shrink rapidly. This premature separation reduced the adhesive

force. In contrast, the pedestal design delayed crack formation by localizing the stress at the center of the pad. This unique characteristic results in a significantly stronger adhesive force because delamination occurs later in the process, allowing the material to remain securely attached to the stamp during the pick-up phase. The post width was optimized using FEM. The graph shows how the energy release rate (y-axis) for the internal cracks (blue curve) and edge cracks (red curve) varies with the post width (x-axis) (Figure 3(j)). And the distribution of normal stress at the interface between the contact pad and the silicon is shown for post widths of 50 μm (left) and 60 μm (right), respectively (Figure 3(k)). At a post width of 50 μm , the internal crack energy release rate is higher than that of edge cracks, meaning stress is concentrated at the center, preventing premature delamination. However, when the post width exceeds 60 μm , the energy release rate for edge cracks surpasses that of internal cracks, indicating that stress at the pad's perimeter initiates delamination. These results suggest that a 50 μm post width is optimal for enhancing adhesion by minimizing edge crack formation. This optimization is based on fixed structural and material parameters, including post height (50 μm), pad dimensions (100 $\mu\text{m} \times 100 \mu\text{m} \times 15 \mu\text{m}$). If these parameters change, the optimal post width may also change, thus requiring further adjustments through FEM.

Figure 3(l) shows the pull-off force versus pulling speed for different post widths—50 μm (green circle), 60 μm (blue triangle), 80 μm (red triangle), and 100 μm (black square). For posts 60 μm and wider, adhesion forces remain very low, with only slight increases as speed rises, similar to flat stamps. In contrast, the 50 μm post shows a distinct behavior, achieving pull-off forces up to 15 mN at low pulling speeds, which decrease as speed increases. This suggests that crack initiation, driven by the structure, plays a more significant role than vdW forces, leading to a $15\times$ improvement in the adhesion performance.

The pedestal structure enhances adhesion during pick-up, but poses challenges during printing owing to the strong adhesive force, necessitating a specialized technique to reduce adhesion. To address this, a laser-assisted thermal expansion technique applies infrared laser pulses (805 nm, 0.2 Hz, 2.3 ms pulse width, 38 mW peak power) to induce localized heating at the interface between the stamp and the micro-LEDs. This creates differential thermal expansion, which weakens the adhesion. Experimental results demonstrate the successful transfer of 100 $\mu\text{m} \times 100 \mu\text{m}$ Si platelets (10 μm thick) and InGaN micro-LEDs (5 μm thick) onto a silicon substrate. By integrating this laser-assisted technique, the pedestal-shaped elastomeric stamp improved the adhesion during pick-up and enabled precise material release during printing. This study shows that the pedestal-shaped stamp is highly effective in maximizing pick-up adhesion, which is critical for improving transfer yield. Although printing is not feasible with this structure alone, its integration with a laser-assisted release process demonstrates a hybrid strategy that addresses trade-offs and ultimately contributes to optimizing overall yield.

2.4. Elastomeric stamp with reservoirs

In 2012, an elastomeric stamp with embedded reservoirs (small chambers for holding pressurized fluid) was developed^[199]. This advanced technology enables the active programmable deformation of the elastomeric surface by controlling the pressure within the reservoirs, thereby allowing for tunable adhesion. This procedure effectively demonstrates the core concept of this technology (Figure 3(m)). During the transfer process, the deflated stamp initially picks up ink. Then, as pressure is applied to the reservoirs, the stamp creates a slight bulge (gentle outward expansion), which reduces the contact area and lowers the adhesion at specific points. This controlled modulation of adhesion improves the precision and efficiency of printing because contact area adjustments allow for accurate material transfer.

The fabrication involves creating an active stamp and thin capping membrane separately, followed by bonding them together (Figure 3(n)). The active stamp consists of a pressurizable reservoir, 225 μm in diameter and 70 μm deep, connected to an external pressurization source via 50 μm -wide microchannels. A thin ($\sim 30 \mu\text{m}$), flexible PDMS membrane covers the reservoir and channels. It inflates into a curved profile under applied pressure to reduce surface adhesion.

The study identifies a reservoir depth of 70 μm in the elastomeric stamp as optimal for effective adhesion control, with varying depths of 30 μm (black line), 70 μm (red line), and 110 μm (blue line) showing distinct impacts on adhesion force (y-axis) as inflation pressure (x-axis) increases (Figure 3(o)). At 70 μm depth, inflation pressure efficiently reduces adhesion within a controlled range, enabling reliable release without full disengagement from the surface. At approximately 10 psi, substantial adhesion reduction was achieved, allowing stable material release in transfer-printing applications. In contrast, the 30 μm depth is too shallow for sufficient inflation, causing minimal membrane deformation under pressure. This results in inefficient adhesion control, in which the adhesion remains too strong for reliable release, making it difficult to detach the materials predictably. Without adequate deformation, the stamp fails to effectively reduce the adhesion, leading to inconsistent transfer and unstable printing performance. On the other hand, the 110 μm depth causes overinflation, where excessive membrane expansion disrupts uniform adhesion control. Consequently, some areas lose contact too quickly, whereas others remain attached, creating an uneven detachment. The adhesion instability caused by overinflation leads to unpredictable release, making the transfer process inconsistent and reducing reliability over repeated cycles. In contrast, the 70 μm depth maintains structural stability under varying pressures, preventing performance degradation from excessive inflation. The active elastomeric stamp enables transfer onto complex surfaces by dynamically modulating adhesion. It successfully prints silicon plates (250 $\mu\text{m} \times 250 \mu\text{m} \times 3 \mu\text{m}$ and 200 $\mu\text{m} \times 200 \mu\text{m} \times 3 \mu\text{m}$) onto flexible PET sheets, which can be bent around curved surfaces (20 mm radius), photonic crystal arrays of SiO₂ microspheres (600 nm radius),

and biological surfaces like leaves, conforming to nonuniform venation. Additionally, a programmable multi-reservoir system consisting of a 5×5 array of reservoirs allows independent actuation of different regions, enabling selective and large-area printing beyond fixed donor layouts. This design enhances scalability, adaptability, and precision, making it highly effective for bioelectronics, sensors, and optoelectronics, where conventional passive stamps fail to achieve conformal contact and selective transfer. This study demonstrates that active control of adhesion via pressurized reservoirs enables precise and selective material release, which is essential for improving transfer reliability. Although the transfer yield was not reported, the ability to dynamically modulate adhesion across complex and non-planar surfaces suggests strong potential for yield enhancement in advanced and flexible applications.

2.5. Shear-enhanced elastomeric stamp

In 2011, the shear-enhanced transfer printing technique was developed, introducing a novel approach to assembly^[200]. Unlike other advanced transfer printing methods, which often rely on specialized structural designs like microtip patterns, pedestal structures, or embedded reservoirs, this technique instead utilizes a conventional flat stamp structure. Rather than modifying the stamp's physical form, it uniquely controls adhesion through precise movement adjustments and applied shear forces. By applying these shear forces, this method reduces adhesion between the flat stamp and the ink, facilitating accurate material transfer to various substrates. The procedures of this technique effectively demonstrate the concept of shear-assisted control, illustrating how shear forces modulate adhesion in a precise manner (Figure 3(p)). In the retrieval phase, the stamp first contacts the donor substrate to pick-up the ink, followed by a rapid retraction to lift the material. In the printing phase, the stamp aligns with the receiver substrate and applies a lateral displacement ($12.5 \mu\text{m}$, corresponding to a shear strain of about 14%) parallel to the substrate surface. This controlled, shear-assisted release gently and accurately places ink by slightly shifting the stamp laterally, uniformly reducing adhesion to prevent sudden detachment and misalignment. This approach effectively achieves delicate material transfers with high precision, without requiring any structural modifications to the stamp.

The elastomeric stamp for shear-enhanced transfer printing was fabricated with a single PDMS post attached to a $950 \mu\text{m}$ -thick backing layer, which stabilized shear-assisted adhesion control. Post dimensions ranged from $100 \mu\text{m}$ to $250 \mu\text{m}$ wide and $50 \mu\text{m}$ tall and can be appropriately selected based on the size of the ink to achieve effective adhesion modulation. Mounted on motorized stages, these stamps controlled lateral and vertical forces, enabling precise material transfer through shear.

The graph illustrates the relationship between shear strain and normalized pull-off force P , where F is the measured pull-off force, E is the Young's modulus, and L represents the post dimensions (Figure 3(q)). The x -axis shows shear strain (%), and the y -axis displays the normalized pull-off force. Data

points indicate different post width sizes: $100 \mu\text{m}$ (square), $150 \mu\text{m}$ (circle), $200 \mu\text{m}$ (green triangle), and $250 \mu\text{m}$ (blue triangle). As shear strain increases, the normalized pull-off force decreases significantly, showing that higher shear strains effectively reduce adhesion between the stamp and the ink. This trend is consistent across post sizes, indicating that shear strain can effectively modulate adhesion in a scalable way. At higher shear strains (12%–16%), the normalized pull-off force nears zero, signifying almost complete detachment.

This graph illustrates across the interface of a stamp-ink system under different shear strains (Figure 3(r)). The x -axis represents the distance along the interface, while the y -axis shows the normal stress in kilopascals (kPa). The preload applied is set at 1.8 mN , ensuring initial contact before the application of shear. At 0% shear strain (black line), the normal stress distribution is relatively uniform across the interface, with minimal stress concentrations. As shear strain increases (represented by different lines: red for 7.4%, blue for 11.1%, and green for 14.8%), significant changes appear at the edges, particularly at the trailing edge (right side of the graph). Higher shear strains lead to increased stress concentration at this trailing edge, while the normal stress along most of the interface remains unchanged. This pattern highlights the mechanism behind shear-assisted adhesion control in transfer printing. By concentrating stress at the trailing edge, the system reduces the overall adhesion across the interface, facilitating easier separation. The localized stress accumulation allows the stamp to detach smoothly from the substrate by focusing the force where it is most effective for controlled release. This controlled shear-strain effect is crucial for minimizing the pull-off force required, as observed in the earlier graphs, and enables precision transfer without needing extensive structural modifications to the stamp itself.

The effectiveness of the shear-enhanced transfer printing technique is demonstrated by its substantial impact on the yield of silicon plates successfully transferred onto a bare substrate under varying shear strain conditions (Figure 3(s)). In this experimental setup, silicon plates with precise dimensions of $100 \mu\text{m} \times 100 \mu\text{m} \times 3 \mu\text{m}$ are subjected to different levels of applied shear strain during the printing process to test yield performance. Results from these tests show a significant increase in yield, from approximately 10% at 0% shear strain to around 95% at an optimal shear strain of about 14%, representing nearly a tenfold improvement. This improvement highlights the method's capability to achieve high-precision and high-yield transfer, even on substrates that generally pose adhesion challenges. This innovative approach broadens the potential applications of transfer printing by enabling highly reliable adhesion modulation through applied shear force, thus facilitating the precise placement of materials onto challenging substrates without the need for additional adhesives.

The principle of controlling adhesion through directional movement, as seen in shear-enhanced techniques, applies in roll-to-roll. By adjusting adhesion based on movement direction, roll-to-roll techniques enable selective, continuous adhesion suited for large-scale production. This approach allows roll-to-roll printing to achieve efficient, high-yield processing across various substrates in an automated process, expanding

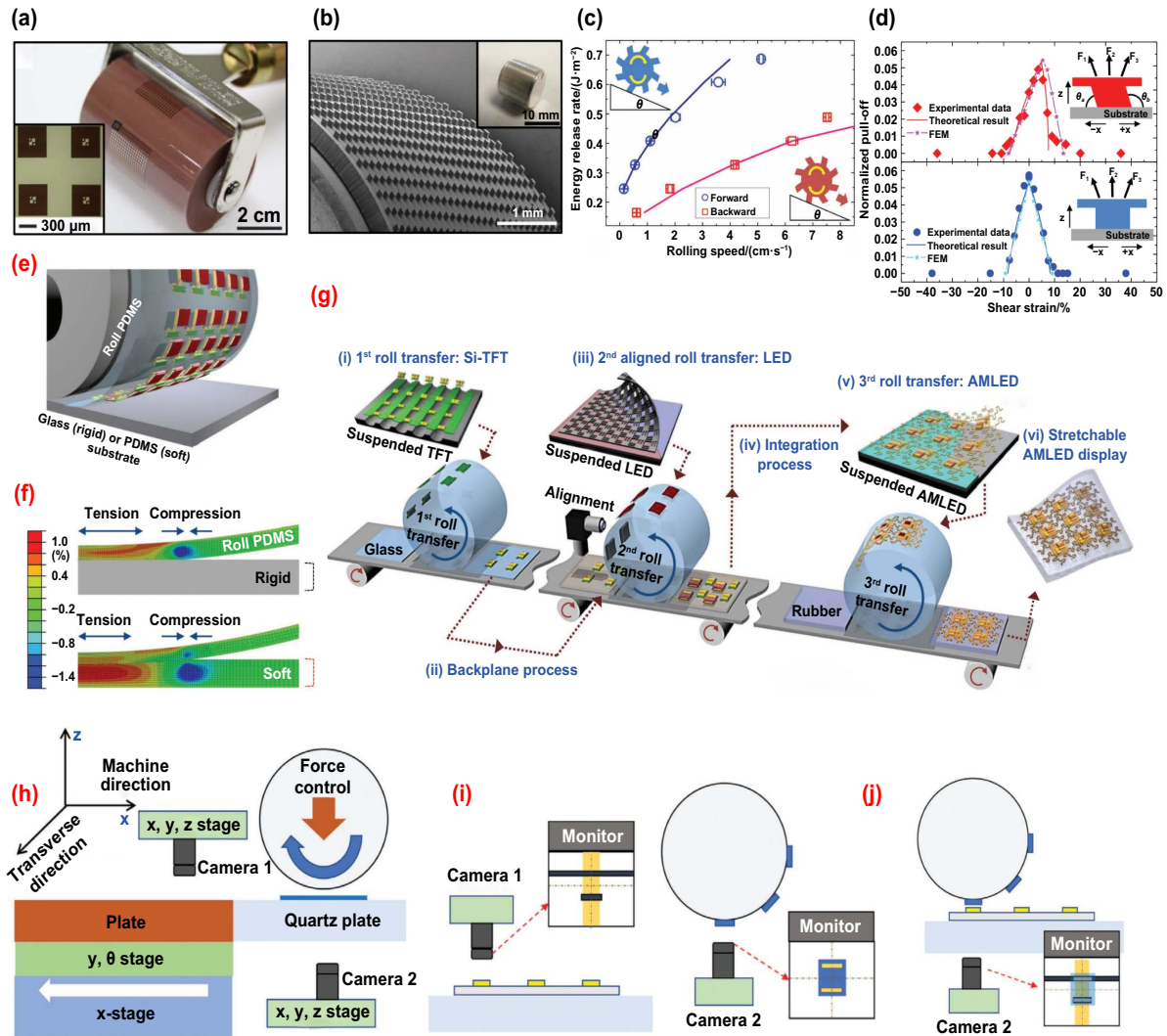


Figure 4. Roll-to-roll transfer techniques. (a) Cylindrical PDMS roll stamp enabling transfer of thin devices via vdW forces. [206] John Wiley & Sons. Copyright © 2011 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim (b) Schematic of a uniquely structured PDMS stamp tilted at 17° . (c) Pull-off force graph comparing adhesion strength in forward (blue) and backward (red) directions. (d) Relationship between shear strain variation and pull-off force according to the angled post structure of the stamp and movement direction. [207] John Wiley & Sons. Copyright © 2012 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim (e) Schematic showing the transfer of devices from a roll-type PDMS stamp to either a glass or PDMS substrate. (f) FEM simulation results depicting deformation behavior of soft and hard substrates under roll-nip pressure. (g) Process flow diagram illustrating the three-step continuous roll-based transfer of Si-TFTs, micro-LEDs, and final display onto the target substrate. [160] John Wiley & Sons. © 2017 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim (h) R2P alignment system capable of sub- $3\ \mu\text{m}$ precision using dual-camera optical feedback and multi-axis control. (i) Illustration of the CA method, where the upper camera measures TCB electrode positions and the lower camera compensates for alignment errors between micro-LED electrodes on the roll and the TCB. (j) RDOA technique utilizing the lower camera to directly align micro-LED electrodes on the roll stamp with those on a transparent electrode substrate. Reproduced from [208]. CC BY 4.0.

the potential of transfer printing for applications requiring precision and scalability.

3. Roll-to-roll based techniques

Micro-LED transfer technology using PDMS stamps is a key method for fabricating high-performance, large-area flexible displays. Conventional transfer printing employs flat PDMS stamps; however, roll-transfer printing using curved PDMS stamps has emerged as an efficient and low-cost alternative for

mass production [206,209–213]. The roll-to-roll transfer method improves the transfer yield through controlled adhesion and precise alignment, enabling high-speed processing of up to 40 000 LEDs per second. However, achieving a high transfer yield requires the optimization of the load pressure, shear strain, and alignment accuracy to prevent device misalignment and ensure stable integration.

Roll-transfer printing modulates adhesion forces through shear strain distribution, allowing the selective detachment of transferred devices. One approach involves shaping the PDMS into a cylindrical roll, enabling vdW force-based

transfer (Figure 4(a))^[206]. This method ensures uniform contact and a high transfer yield. Another approach utilizes a uniquely structured PDMS stamp with a 17° tilt to control adhesion (Figure 4(b))^[207]. The tilted structure induced shear strain variations depending on the direction of movement (Figure 4(c)). In the forward direction (F_1), force is applied along the post tilt, suppressing crack propagation and increasing adhesion, enhancing pick-up efficiency. In contrast, in the backward direction (F_3), force is applied against the post tilt, accelerating crack propagation and reducing adhesion, improving release efficiency (Figure 4(d)). This controlled adhesion behavior allows precise pick-up and placement, thereby enhancing transfer reliability.

This three-step roll-transfer process enables high-precision micro-LED integration. Si-thin film transistors (TFT) were fabricated on silicon on insulator (SOI) wafers via ion implantation and thermal oxidation to form gate insulators. To maintain alignment, the TFTs are suspended and anchored at both edges, supported by a buried oxide (BOX) layer that protects the active Si layer from trapped charges and surface roughness^[160]. Following Si-TFT transfer, micro-LEDs are picked up from an AlInGaP wafer with an AlAs sacrificial layer, which is selectively etched to release suspended micro-LEDs. These devices remained stabilized by bridge structures facilitating high-yield pick-up and precise placement using roll stamps. In the final integration step, the components are transferred to a stretchable elastomer or bendable plastic substrate. Throughout the process, optical alignment feedback ensures submicron precision.

Two major challenges in roll-transfer printing are roll-nip pressure control and precise synchronization of roll motion and substrate translation. Maintaining a uniform roll-nip pressure prevents deformation-induced misalignment, as FEM simulations indicate that hard substrates improve alignment accuracy, whereas soft substrates enhance stretchability but increase lateral deformation (Figures 4(e) and (f)). To achieve precise alignment, synchronization between the angular displacement of the roll and the translational movement of the receiver substrate is necessary (Figure 4(h))^[208]. Calibration alignment (CA) (Figure 4(i)) and roll-based direct overlay alignment (RDOA) techniques (Figure 4(j)) improve placement accuracy, reducing misalignment to below 1 μm . These optimizations enhanced the transfer yield, approaching 99.9%, while ensuring device integration accuracy. The roll transfer process employs contact mechanics and adhesion modulation to achieve precise device placement. The cylindrical PDMS stamp enabled controlled pick-up and release through the shear strain distribution across the roll nip. FEM simulation under a nip force of $2.8 \text{ N}\cdot\text{mm}^{-1}$ replicating actual roll-transfer conditions on a unit pixel shows that less than 0.1% tensile strain occurs during the process, ensuring mechanical reliability and enabling process optimization.

The roll-transfer process exhibits superior performance in terms of yield, alignment precision, and scalability. The optimized PDMS roll stamps achieved a transfer yield of nearly 100%, surpassing that of traditional pick-and-place methods. Automated roll-to-plate (R2P) transfer machines maintain alignment errors within 3 μm , while RDOA further

improves precision to 0.52 μm in the machine direction (MD) and 0.40 μm in the transverse direction (TD), outperforming CA, which achieves 2.22 μm MD and 2.59 μm TD accuracy (Figure 4(g)). In terms of production speed, conventional transfer methods operate at 2–10 LEDs per second, whereas roll-transfer technology achieves 40 000 LEDs per second, making it suitable for large-area micro-LED manufacturing. The roll-transfer method offers high-speed, scalability, and high-precision integration for next-generation display fabrication. This process supports mass production and significantly improves throughput. Their compatibility with flexible and stretchable substrates has facilitated advancements in wearable electronics and deformable optoelectronic devices. The submicron alignment accuracy ensures precise device placement and minimizes defects.

However, challenges remain in maintaining roll-nip pressure uniformity because fluctuations may cause localized misalignment. In addition, although soft substrates improve flexibility, they introduce positioning accuracy issues that require further refinement. Despite these limitations, roll-transfer printing represents a transformative advancement in flexible optoelectronics, bridging the gap between traditional semiconductor processing and high-throughput micro-LED manufacturing. This paper presents roll-transfer printing as a high-precision, scalable technique for micro-LED and Si-TFT integration that enables the mass production of stretchable electronic displays. By optimizing the roll-nip pressure, alignment synchronization, and adhesion modulation, this method achieved a near-perfect transfer yield and submicron precision. Experimental results confirm its feasibility for high-throughput manufacturing, achieving 40 000 LEDs per second with alignment errors below 1 μm . Future research will focus on enhancing the mechanical stability, improving automated alignment, and expanding substrate adaptability to further advance scalable flexible optoelectronics. This technique is meaningful in that it shifts the discussion of transfer yield toward high-throughput scenarios, where large numbers of micro-LEDs must be transferred in a short time. Achieving perfect yield with only a few devices over a long period may not be commercially viable, as it compromises cost-efficiency. Roll-to-roll printing opens the possibility of maintaining high yield while enabling rapid, large-scale transfer, which is essential for scalable manufacturing.

4. Electrostatic and electromagnetic based techniques

4.1. Electrostatic

The electrostatic-based mass-transfer method developed by LuxVue (later acquired by Apple Inc.) offers a precise and efficient approach for micro-LED transfer^[214–220]. This method uses electrostatic forces to transfer micro-LEDs from a donor substrate to a receiver substrate by switching between attraction and repulsion, as shown in Figures 5(a) and (b). At the core of this technique is an electrostatic transfer head array composed of an electrode layer and a dielectric layer, which are structured into either mesa formations or convex pillars.

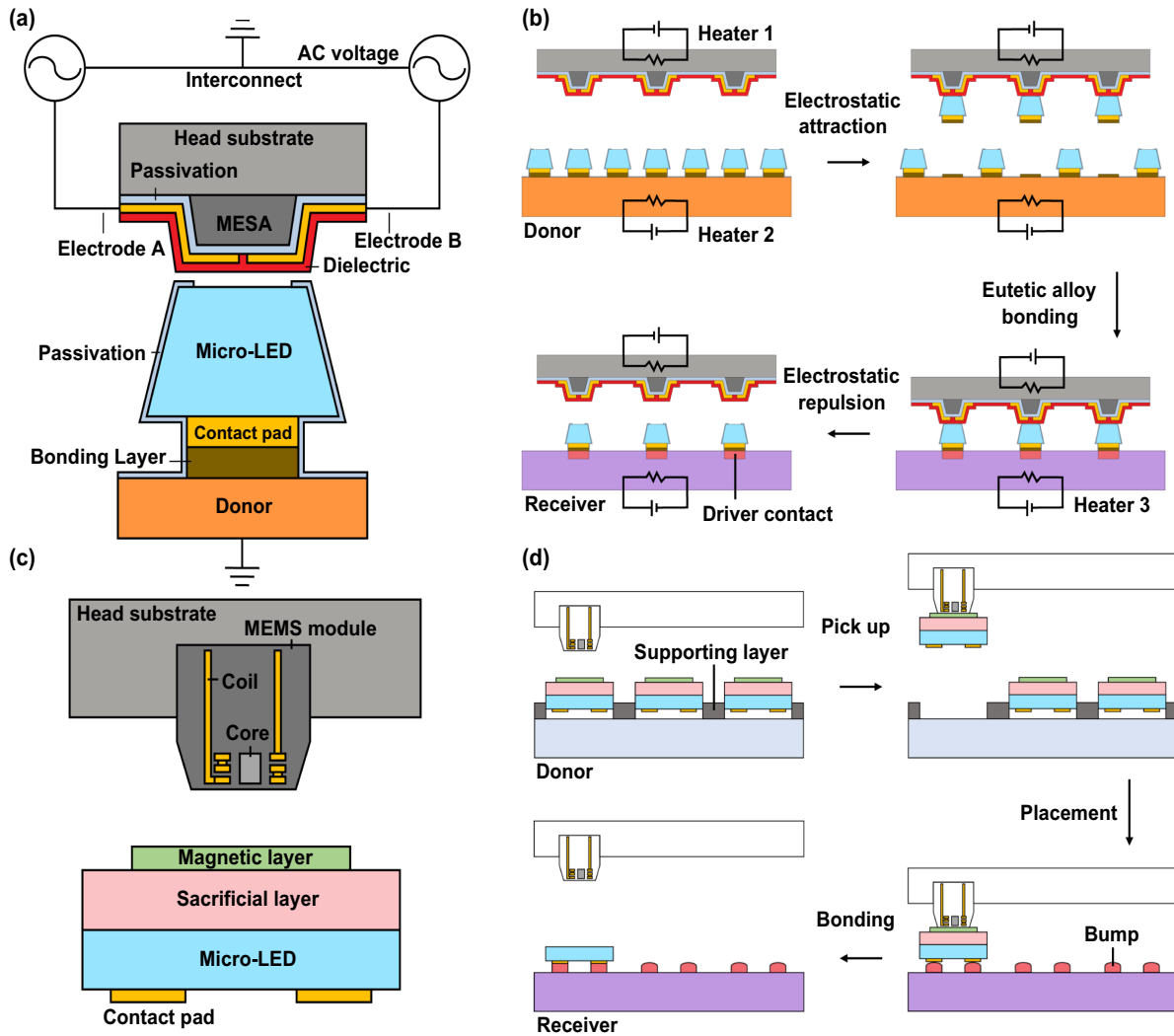


Figure 5. Electrostatic and electromagnetic-based technique. (a) Electrostatic transfer head consisting of passivation layer, electrode layer, and dielectric layer designed as mesa structure on transfer head substrate, which is positioned over micro-LED with a certain gap. (b) Transfer printing process using the electrostatic pick-up technique. (c) Electromagnetic transfer head consisting of MEMS module composed of ferromagnetic metal between electromagnetic coils, which is aligned over micro-LED coated with magnetic layer and sacrificial layer on the surface of the micro-LED. (d) Transfer printing process using electromagnetic pick-up technique.

From a yield perspective, the key challenge in this technique lies in preventing LED damage caused by electrostatic forces, making precise control of voltage and device structure essential for maintaining high transfer yield. This design ensured that the pitch of the electrostatic heads matched an integer multiple of the micro-LED array pitch, thereby enabling a precise alignment and damage-free transfer (Figure 5(a)). Such a configuration promotes high alignment accuracy while minimizing the risk of damage during the transfer process, which begins by positioning the electrostatic heads above the micro-LED array on the donor substrate with a carefully maintained gap. When a voltage is applied, the electrostatic heads create an attractive force that selectively picks up the target micro-LEDs, while avoiding the unintentional displacement of nearby units. Heater 1 then warms the transfer heads, allowing heat to flow to the micro-LEDs, while heater 2 heats the first bonding layer, stabilizing the bonding interface and preparing it for attachment. Once the micro-LEDs are picked up,

they are moved to the receiver substrate, where a second bonding layer is prepared. Eutectic bonding, facilitated by heater 3 on the contact pads, provides a stable, low-resistance connection that is essential for the electrical performance of micro-LEDs. This bonding step securely attaches the micro-LEDs to the receiver substrate while ensuring minimal electrical resistance. To release the micro-LEDs onto the receiver substrate, the voltage is turned off, creating a controlled electrostatic repulsion that gently detaches the micro-LEDs from the heads without causing damage. The electrostatic transfer head array then retracts and is ready to begin the next transfer cycle. The key to the success of this method is the dielectric layer on the transfer heads and careful voltage control, which prevent damage to the micro-LEDs. The forces were adjusted to be strong enough for effective transfer, yet gentle enough to avoid structural stress. This method achieves high transfer rates, reaching several million units per hour, with consistent repeatability, and no residue remains on the substrate. The

efficiency, precision, and residue-free nature of electrostatic-based transfer make it a scalable and dependable choice for large-scale micro-LED assemblies, meeting the stringent accuracy and efficiency demands of advanced micro-display technologies.

4.2. Electromagnetic

The electromagnetic-based transfer method developed by the Industrial Technology Research Institute (ITRI) and PlayNitride represents an innovative approach for micro-LED transfer technology^[221]. While similar in concept to the electrostatic transfer method, this technique instead utilizes electromagnetic forces to achieve precise micro-LED placement (Figures 5(c) and (d)). The electromagnetic transfer head was designed with microelectromechanical systems (MEMS) incorporating electromagnetic coils positioned around ferromagnetic metal elements that generate the magnetic fields necessary for the transfer (Figure 5(c)). Before the micro-LEDs can be transferred using electromagnetic forces, they must be prepared and transferred onto a temporary donor substrate. This is accomplished through a laser lift-off (LLO) process that detaches the micro-LEDs from their original substrate and secures them onto the donor substrate. To facilitate magnetic interaction, a sacrificial layer made of organic materials is applied to the surface of the micro-LEDs, followed by the deposition of a thin magnetic coating layer, approximately 1 μm thick. This magnetic layer is typically composed of Ni-based metals or other ferromagnetic materials, enabling the micro-LEDs to interact with the electromagnetic fields generated by the transfer head array. Once the micro-LEDs are ready, the transfer head array, equipped with electric-programmable MEMS modules, is aligned above the micro-LED array on the donor substrate. Electromagnetic attraction is then activated, allowing the transfer head to selectively pick-up the micro-LEDs by overcoming the adhesion forces that hold them to the donor substrate. This process ensures that only the desired micro-LEDs are transferred, thereby enhancing the selectivity and precision of the method (Figure 5(d)). After the micro-LEDs are picked up, they are moved to the receiver substrate, where they are carefully positioned. Upon placement, the micro-LEDs are heated to initiate the formation of a metal–alloy bond, securing them to the receiver substrate. This bonding process provides a stable, low-resistance connection that is essential for the electrical and thermal performance of the micro-LEDs in their final application. It is important to note that the quality and consistency of the magnetic coating layer on the micro-LEDs play a crucial role in the accuracy and repeatability of the electromagnetic transfer process. Variations in the homogeneity of magnetic materials can affect the effectiveness of the transfer; therefore, precise control over the coating layer is essential, as this electromagnetic-based transfer method can achieve a transfer rate of nearly one million units per hour, combining high speed with excellent selectivity. This performance makes it a promising technique for large-scale micro-LED assembly, and provides an efficient and precise solution for high-throughput manufacturing.

5. Laser-based techniques

Laser processing offers several unique advantages, including a high processing speed, precise spatial control with selective irradiation, high material selectivity across multimaterial systems, and large-area processability^[222,227–235]. Because of these features, laser-based techniques have evolved into highly adaptable, material-compatible, and scalable processes. Representative examples include laser dicing, thermal annealing, LLO, laser-assisted transfer and patterning, and laser-assisted die transfer.

Unlike the conventional direct laser irradiation of the target material, these techniques typically rely on laser irradiation through a transparent substrate, where the laser energy is selectively absorbed by an interfacial active layer (Figure 6(a)). This approach triggers a variety of physical and chemical interactions at the interface, enabling the precise control of the interfacial phenomena required for material transfer and processing.

To achieve the desired interfacial phenomena, the laser fluence, pulse duration, and other key laser processing parameters must be carefully optimized to control the thermal effects at the interface. A critical design principle is that the laser wavelength should be selected such that the laser-active layer strongly absorbs the incident laser, while the transparent substrate remains largely transparent. This condition is governed by the band gap discontinuity between the two materials, which can be described by the inequality: $E_{\text{transparent}} > E_{\text{light}} > E_{\text{active}}$, where E_{active} is the band gap of the active layer, $E_{\text{transparent}}$ is the band gap of the substrate, and $E_{\text{light}} = h\nu$ is the photon energy of the incident laser. This bandgap engineering ensures that the active layer efficiently absorbs laser energy while minimizing unnecessary absorption or scattering within the transparent substrate. LED damage during transfer can significantly degrade the overall yield; therefore, minimizing such damage is critical for achieving high transfer efficiency and process reliability. To further prevent potential optical damage to the adjacent layers, the optical penetration depth must be minimized. In this context, short-wavelength UV lasers are preferred because of their naturally shallow penetration depth, typically of the order of a few hundred nanometers. Consequently, even a thin active layer can effectively confine laser interactions and protect nearby structures from unintended optical damage.

Overall, the effectiveness of laser-based transfer and processing techniques critically depends on the careful combination of material system design and laser process parameter optimization tailored to specific applications and target performances.

Laser-based transfer printing technology has emerged as a pivotal solution for high-precision, scalable, and flexible assembly of micro-LEDs^[223]. By minimizing the mechanical stress during the transfer process, this technique enables deterministic and accurate placement of micro-LEDs onto various receiver substrates. Compared to conventional pick-and-place methods, laser-based transfer techniques allow for high-throughput selective transfer, while accommodating a wide range of micro-LED sizes and substrate materials, thus

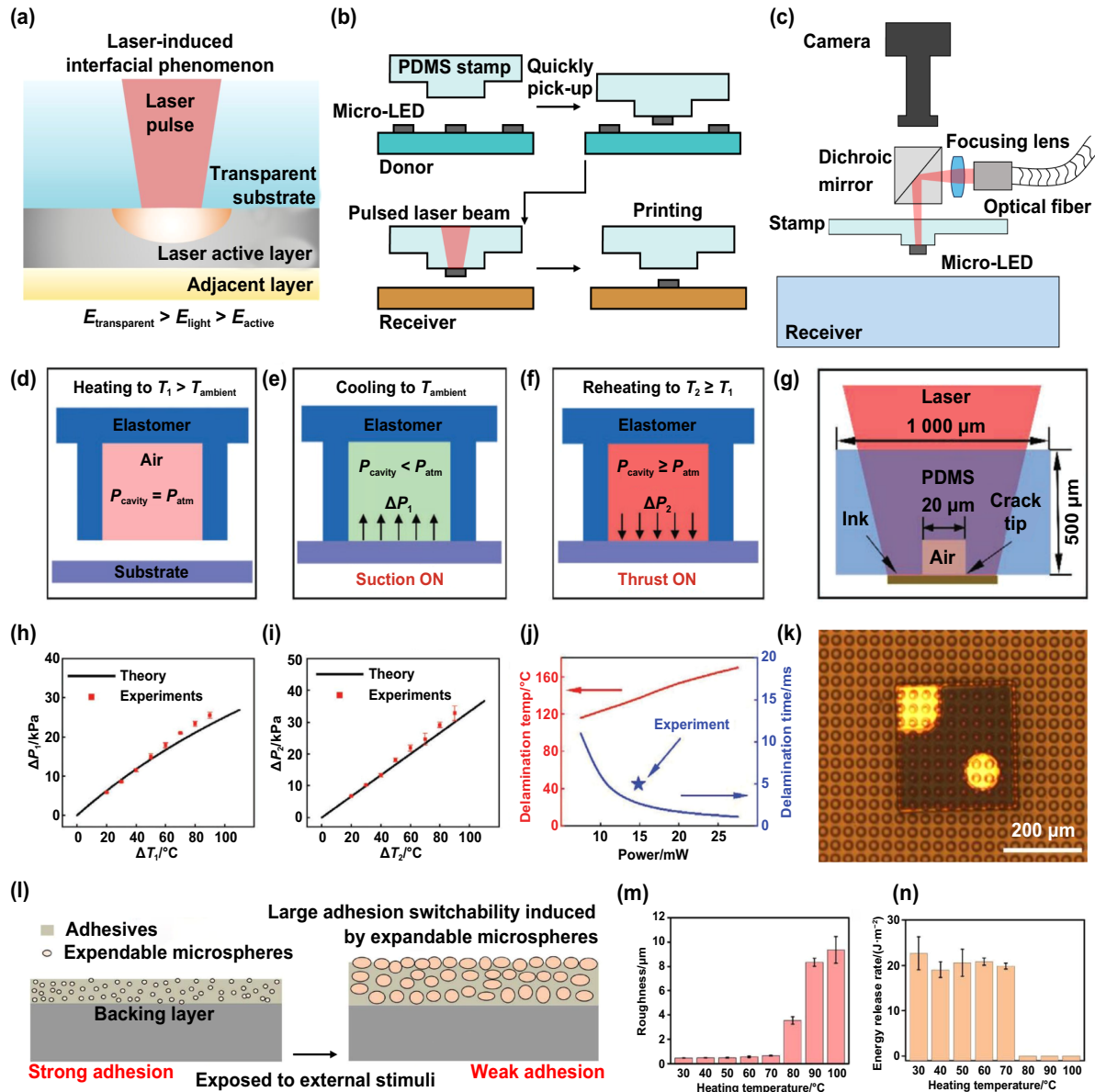


Figure 6. Laser-based techniques. (a) Laser-induced interfacial phenomenon. [222] John Wiley & Sons. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (b) Schematic of the LMTP steps using an elastomeric stamp. (c) A schematic view of the LMTP setup. (d)–(f) Laser-driven transfer printing using a thermally tunable stamp. (g) The geometric characteristics of the elastomer stamp. (h)–(j) Dependence of delamination temperature and time on absorbed laser power. (k) Transfer printing of micro-LEDs onto a PDMS stamp. [226] John Wiley & Sons. © 2021 Wiley-VCH GmbH. (l) Schematic of a thin sheet stamp integrating a thermally expandable microsphere adhesive layer for transfer printing. (m) Effect of heating temperature on the surface roughness of the stamp. (n) Energy release rate between the stamp and glass at different heating temperatures. Reproduced with permission from [161]. Copyright © 2020 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC).

offering superior process flexibility. Recent advancements in both contact and non-contact laser transfer techniques have significantly enhanced transfer efficiency, alignment precision, and process yield, paving the way for scalable manufacturing of large-area micro-LED displays. Each laser-based transfer printing method has unique strengths and operational characteristics, making them highly versatile for a broad spectrum of micro-LED assembly processes.

The laser-driven micro transfer placement (LMTP) process integrates the key advantage of conventional transfer

printing—the direct, high-precision, and selective pick-up of micro-LEDs from the growth or fabrication substrate—with laser-assisted release for accurate placement (Figure 6(b)). The micro-LED pick-up process is similar to conventional transfer printing techniques that utilize elastomeric stamps. In this process, a PDMS stamp is first brought into contact with the donor substrate to facilitate the rapid retrieval of the micro-LEDs. Upon transferring the stamp to the receiver substrate, the stamp–micro-LED interface was precisely irradiated with a laser pulse, triggering the controlled release

of the micro-LEDs and enabling their accurate placement onto the receiving substrate. This method utilizes a significant thermal-mechanical mismatch at the stamp/micro-LED interface to induce delamination. The micro-LED pick-up process is similar to conventional transfer-printing techniques that employ elastomeric stamps such as PDMS. In this approach, the micro-LEDs were picked from the donor substrate using a PDMS stamp, after which a pulsed laser beam (power: 30 W, wavelength: 805 nm, minimum pulse width: 1 ms) is directed onto the upper stamp surface (Figure 6(c))^[224]. The effectiveness of this non-contact laser transfer printing technique is highly dependent on the characteristics of the laser pulses employed during the process. The laser beam is transmitted through the transparent PDMS stamp, whereupon the micro-LED, which possesses a high absorption coefficient for the incident laser, undergoes localized heating. Once the micro-LED absorbs thermal energy, it conducts heat to the relatively cooler stamp, elevating the interface temperature to 250 °C–300 °C. As the contact interface heats, the thermal expansion coefficient mismatch between the stamp and the micro-LED, combined with the mechanical constraints at the interface, induces thermal expansion and interfacial stress. When the released energy exceeds the adhesive energy, the micro-LED detaches from the PDMS stamp, enabling controlled release and high-precision transfer. The LMTP process provides a high-precision, non-contact approach for micro-LED transfer, addressing scalability and placement accuracy challenges in advanced electronic manufacturing systems. Although it offers deterministic release and broad substrate compatibility, excessive thermal energy can increase the strain energy release rate, potentially leading to microstructural damage, which remains a key limitation of the process.

The thermally controlled tunable stamp mechanism enables the dynamic modulation of interfacial adhesion for both microstructure retrieval and deterministic release during transfer printing (Figures 6(d)–(f))^[225,226]. This approach enhances adhesion during pick-up while ensuring precise and controlled release during printing, thereby improving process reliability and scalability. Upon heating the stamp, which incorporates an air-filled cavity embedded within an elastomeric surface, to a temperature exceeding the ambient temperature (T_{ambient}), the internal air pressure within the cavity equalizes to the atmospheric pressure ($P_{\text{cavity}} = P_{\text{atm}}$) (Figure 6(d)). In this state, the stamp is in contact with the target substrate. Subsequent cooling to ambient temperature induces contraction of the cavity air, resulting in a negative pressure ($P_{\text{cavity}} < P_{\text{atm}}$) that generates a strong suction force ($\Delta P_1 = P_{\text{atm}} - P_{\text{cavity}}$) (Figure 6(e)). Conversely, reheating the stamp to a temperature T_2 above T_1 increases the internal air temperature, leading to a positive pressure generation ($P_{\text{cavity}} > P_{\text{atm}}$). This results in a thrust force ($\Delta P_2 = P_{\text{cavity}} - P_{\text{atm}}$) that facilitates the controlled detachment of the microstructure from the stamp and its accurate placement onto the receiving substrate (Figure 6(f)).

Building on this thermally controlled tunable stamp mechanism, the fundamental relationship between temperature variation and interfacial pressure modulation was theoretically modeled and experimentally validated to ensure precise and repeatable transfer performance. This thermally tunable

mechanism provides active control over suction and thrust activation (suction ON and thrust ON), ensuring deterministic modulation of interfacial adhesion and thereby enhancing the precision and reliability of the transfer printing process (Figure 6(g)).

Figure 6(h) presents the variation in suction pressure (ΔP_1) as a function of temperature decrease (ΔT_1). As the temperature decreases from 120 °C to 30 °C, the internal cavity pressure reduces, thereby increasing the suction force. Similarly, Figure 6(i) illustrates the dependence of thrust pressure (ΔP_2) on temperature increase (ΔT_2). When the temperature rises from 30 °C to 120 °C, the air temperature within the cavity increases, generating positive pressure, which subsequently results in a proportional increase in thrust force. A greater temperature differential enhances both the suction and thrust forces, confirming the strong dependence of interfacial adhesion modulation on thermal variation. Despite the nonuniform temperature distribution inside the cavity, the overall temperature increase effectively increases the internal pressure, facilitating the smooth release of the micro-LEDs from the stamp. The release time does not exceed 10 ms, while the release temperature remains below 165 °C (Figure 6(j)). These results demonstrate that high-speed transfer printing can be achieved without inducing thermal damage at the interface. The optical image presents that micro-LEDs were successfully transferred using the tunable stamp (Figure 6(k)). This thermally controlled adhesion mechanism, particularly at the microscale, plays a critical role in enhancing the reliability of both pick-up and release processes.

Technology that combines the advantages of laser-based transfer printing with a programmable and scalable approach enables the cost-effective fabrication of micro-LED arrays while ensuring high reliability and efficiency^[161]. Figure 6(l) shows a schematic illustration of the design of a thin-sheet stamp that incorporates a thermally expandable microsphere adhesive layer for transfer printing. Upon exposure to external stimuli, the substantial volumetric expansion of the microspheres effectively induces a hierarchical microstructure on the surface. In the initial stage, the small size of the embedded expandable microspheres rendered their impact on the adhesion strength of the adhesive layer on flat terrain negligible. However, exposure to external thermal stimuli at 90 °C results in the rapid volumetric expansion of the microspheres, effectively inducing a hierarchical microstructure on the surface of the adhesive layer, which significantly reduces adhesion strength. This facilitated the successful release of chips attached to various receiver substrates.

The hierarchical microstructure formed on the stamp surface was confirmed by SEM images taken before and after the application of the thermal stimulus at 90 °C. A quantitative study of the surface roughness (R_a) and energy release rate (adhesion strength) of the thermal release tape (TRT) stamp across various temperatures revealed a significant change at 80 °C, indicating a liquid-vapor phase transition (Figures 6(m) and (n)). The similar temperature dependence of surface roughness and adhesion strength demonstrates that the reduction in adhesion at 90 °C is attributed to a substantial increase in surface roughness, underscoring the importance of surface

microstructure in adhesion control. The exceptional adhesive transition of the stamp surpasses that of conventional elastic stamps with predefined surface-relief microstructures, enabling effective ink transfer and release on a variety of substrates. A laser-based programmable transfer-printing leverages the rapid adhesive transition of stamps in response to external thermal stimuli, integrating the advantages of a digitally controlled laser-heating system. The apparatus consists of a digitally controlled laser system designed to induce a weak adhesive state in the stamp, an optical microscope for in situ monitoring, and an automated conversion stage for precise positioning and alignment of the ink and the laser. In programmable transfer printing, the array of ink (micro-LEDs) manufactured on the source wafer is first picked up by the stamp. Subsequently, the adhesion of the individual ink to be transferred is locally irradiated and released, enabling the selective printing of the desired pattern onto various receiver substrates. This capability allows the release of small, ultrathin, and delicate functional components, which play a crucial role in the cost-effective mass production of wearable devices.

6. Vacuum-based techniques

Vacuum-assisted pick-and-place (VAPP) technology has been widely used for transferring microelectronic components, including LED chips, because of its efficiency and simplicity^[236]. This method utilizes a pressure differential to generate suction, which enables the components to be lifted from a donor substrate and placed onto a receiver substrate (Figures 7(a) and (b)). While effective for larger components, VAPP struggles with micro-LEDs owing to the reduced contact areas and increased vdW forces, leading to transfer failures and misalignment. Furthermore, the mechanical stress from vacuum suction can cause deformation or damage to delicate components, limiting its application in high-precision microdevice assemblies.

To address these challenges, the microvacuum-assisted selective transfer (μ VAST) method was developed (Figure 7(c)) by integrating a vacuum controllable module (VCM) with microhole arrays and PDMS microchannels^[162]. This system enables precise, non-invasive transfer by dynamically modulating the adhesion forces. Adhesion switchability is a key parameter that ensures optimal pick-up and release, reduces mechanical stress, and improves yield. The ability to selectively engage vacuum suction in microscale regions allows for highly controlled handling of micro-LEDs, minimizing unintended adhesion or detachment.

The μ VAST mechanism relies on controlled vacuum suction through microengineered structures. The VCM, which consists of microhole arrays on a glass substrate with hard PDMS microchannels, enables fine-tuned adhesion modulation. Balancing vdW interactions and vacuum suction allows for stable yet gentle handling of micro-LEDs, minimizing damage and improving placement accuracy. The integration of SU-8 micropillars around the microholes enhanced the adhesion force by reducing air leakage, further optimizing the control over the attachment and detachment cycles.

A schematic representation of the μ VAST transfer process is shown (Figure 7(d)). Initially, a micro-LED is aligned with the center of the microhole and vacuum suction is applied to establish secure adhesion. Once the micro-LED detaches from the donor wafer, it remains held in place until it reaches the target substrate. Upon vacuum deactivation, the micro-LED is precisely released onto the new surface without mechanical damage (Figure 7(e)).

The fabrication of the μ VAST system involves laser-induced etching (LIE) to create uniform 20 μ m microholes in a glass substrate (Figure 7(f)). Femtosecond infrared (IR) laser pulses induce localized structural modifications and defects that accelerate wet etching. This approach ensures precise microhole formation with minimal surface damage. SU-8 photoresist is then patterned to form micropillar arrays around the microholes, whereas hard PDMS structures are fabricated to create microchannel networks. These PDMS microchannels are bonded to a glass substrate via oxygen plasma treatment to ensure robust adhesion. The SU-8 micropillars play a critical role in enhancing the transfer efficiency by minimizing the contact area between the VCM and the microchips, thereby reducing the vdW forces during the transfer process (Figure 7(g)). Additionally, the micropillars help maintain structural integrity by preventing the deformation of the PDMS microchannels under vacuum conditions. The combination of precise laser etching and the microstructural design results in a highly efficient and scalable microtransfer platform.

A notable feature of the μ VAST system is its capability to perform both selective and entire transfer processes based on the activation of vacuum channels. During selective transfer, individual vacuum channels within the VCM can be selectively activated, allowing the transfer of micro-LED arrays corresponding to specific channels. For the entire transfer, multiple or all the vacuum channels were engaged concurrently, enabling the transfer of larger micro-LED arrays or full device areas in a single step (Figure 7(h)). This flexible control mechanism enhances the versatility of the μ VAST method, making it adaptable to various application requirements and substrate types.

Another critical aspect of the μ VAST system is its ability to generate sufficient suction force to break the μ -bridge, a microscale connecting structure that temporarily holds microchips (such as micro-LEDs) on the donor wafer. The μ -bridge acts like a thin tether that secures the chip in place until it is ready for transfer. To detach the chip successfully without causing damage, the suction force must exceed the critical fracture force required to break the bridge. μ VAST achieves this by applying a vacuum suction force that is strong enough to cleanly sever the μ -bridge while minimizing mechanical stress on the microchip. This precise control ensures high transfer reliability and maintains the structural integrity of delicate components throughout the pick-up and release processes. This mechanism shows that even with increasing microhole diameters, the suction force remains sufficient to break the μ -bridge with minimal performance degradation (Figure 7(i)). Additionally, the graph shows that the suction force is consistently maintained even as the number of μ s

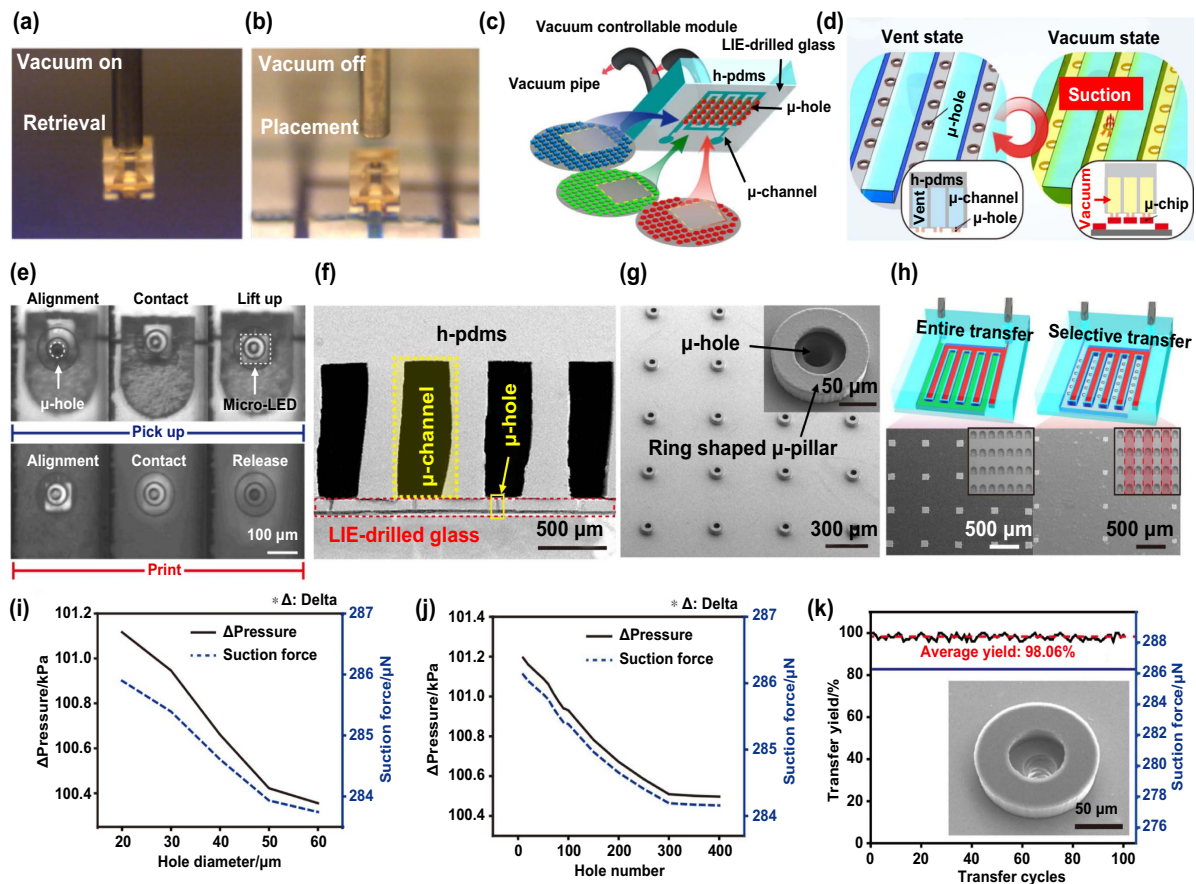


Figure 7. μ VAST system and transfer process. (a) Vacuum-assisted pick-and-place (VAPP) retrieval step and (b) placement step. [236] John Wiley & Sons. © 2017 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (c) μ VAST system integrating a vacuum control module (VCM) with microhole arrays and PDMS microchannels for precise transfer. (d) μ VAST transfer process covering alignment, suction, pick-up, and release of micro-LEDs. (e) Microscopic images of the μ VAST process, showing alignment, contact, pick-up, and release stages. (f) Fabrication of microholes via LIE for creating uniform microhole arrays. (g) Structural design of SU-8 μ -pillars and PDMS microchannels to improve adhesion control and mechanical stability. (h) Selective transfer and entire transfer processes enabled by the independent activation of vacuum channels within the μ VAST system. (i) Relationship between suction force and pressure difference with varying microhole diameters, maintaining sufficient force to break the μ -bridge. (j) Consistent suction force and pressure difference even with increasing numbers of microholes, ensuring stable transfer efficiency. (k) Transfer yield during 100 μ VAST cycles of 10×10 μ LED arrays, achieving a high yield of 98.06%. Reproduced from [162], CC BY 4.0.

increases, ensuring stable transfer efficiency without compromising reliability (Figure 7(j)).

The graph illustrates the average transfer yield during 100 μ VAST cycles of 10×10 μ LED arrays (Figure 7(k)). The μ VAST method achieves an impressively high transfer yield of 98.06%, demonstrating its superior reliability and consistency. This outstanding yield reflects the ability of the system to maintain stable suction forces and precise control over the pick-up and release processes throughout repeated cycles. The SEM images captured after these cycles confirmed that the VCM structure exhibited minimal damage, highlighting its structural robustness and durability. The consistent performance and mechanical stability are attributed to the robust design of the microhole arrays and the uniform distribution of vacuum suction. These quantitative results underscore the effectiveness of μ VAST as a highly reliable technique for scalable, high-yield micro-LED manufacturing.

Importantly, while the current demonstrations primarily focus on micro-LEDs around 50 μ m, the fundamental

principles of μ VAST—precise adhesion control via microengineered vacuum channels—are inherently scalable. By reducing microhole diameters and optimizing microchannel geometry, the technique holds strong potential for handling smaller chips, including those in the 20 μ m or even 10 μ m range.

Beyond conventional techniques, μ VAST enables micro-LED placement on flexible substrates such as PI as well as hornet wing, leaf, fabric, skin, rubber, and paper textiles. Independent vacuum channel control allows for both selective and batch transfers, thereby improving scalability. Challenges remain in achieving higher aspect ratio microholes (>15) for large-area transfer and refining auto-alignment systems. Despite these hurdles, μ VAST represents a significant advancement in micro-LED manufacturing, facilitating next-generation optoelectronic applications. Continued advancements in laser etching and vacuum modulation will further enhance their potential for mass production and commercial adoption.

7. Fluidic self-assembly-based techniques

Fluidic self-assembly (FSA) is a technique used to transfer and assemble micro- or nanoscale components, such as micro-LEDs, onto a target substrate using a fluidic medium. Originally introduced by Yeh and Smith in 1994, this method demonstrated for the first time the integration of GaAs-based LEDs onto silicon substrates via fluid-mediated transport and shape-directed alignment^[238]. Since then, FSA has proven to be an effective means of stabilizing component orientation and positioning, enabling cost-effective, high-throughput, and large-scale assembly. Key principles include the utilization of carrier fluids for component transport and geometric matching for precise positioning to ensure proper assembly alignment and stable integration.

The process begins with the preparation of discrete trapezoidal GaAs LED blocks fabricated using molecular-beam epitaxy (MBE) and temporarily bonded to an intermediate substrate (Figure 8(a)). These blocks are then released into a carrier fluid—typically ethanol—and introduced onto a Si substrate patterned with matching trapezoidal cavities (Figure 8(b)). The geometric design ensures the GaAs blocks align correctly due to gravity and fluid dynamics, settling into the pre-etched Si cavities. Subsequent fluid removal, performed without disturbing the orientation, finalizes the precise integration. In the final assembled state, the GaAs LEDs are securely embedded into the Si substrate and can emit light through defined top apertures (Figure 8(c)).

The fabrication initiates with MBE growth of GaAs structures on AlAs etch-stop layers, forming trapezoidal mesas via ion milling. Subsequent high-temperature wax bonding onto an intermediate substrate facilitates backside metallization. Following selective etching removal of the GaAs substrate, the isolated LED blocks are transferred into the fluidic medium. Simultaneously, the Si substrate undergoes anisotropic wet chemical etching to create complementary trapezoidal cavities. Post-transfer, fluid evaporation is carefully managed to preserve structural integrity and prevent residue formation.

Optimization of fluidic self-assembly involves controlling trapezoidal geometry and etching accuracy to ensure reliable alignment. Crucial parameters include precise control over sidewall angles—typically between 55° and 68°—to guide effective self-alignment. Additional optimization is required in managing fluid evaporation rates using supercritical CO₂ to mitigate surface tension effects and prevent misalignment. Maintaining high purity and cleanliness during fluid processing steps is also vital to prevent contamination that could degrade the optical and electrical performance of assembled LEDs.

The core mechanism exploits shape-recognition, where the trapezoidal geometry dictates selective and orientation-specific placement. Fluidic transport delivers components close to their matching cavities, where gravitational and capillary forces facilitate precise alignment. Self-correction occurs naturally due to the energy minimization principle, as the trapezoidal blocks fit snugly into complementary holes. Fluid evaporation via supercritical drying preserves this alignment,

eliminating detrimental capillary forces that could otherwise disturb the precise arrangement.

Whereas early FSA methods relied on gravitational forces and passive immersion, recent advancements have integrated dielectrophoresis (DEP), magnetic forces, and controlled agitation^[163]. These improvements significantly enhance both the precision and yield, making FSA a viable solution for assembling delicate microstructures such as micro-LEDs (Figure 8(d)).

Magnetic-force-assisted dielectrophoretic self-assembly technology (MDSAT) is an FSA method that integrates magnetic and DEP forces to achieve precise micro-LED alignment. In this approach, micro-LEDs embedded with nickel are manipulated by magnetic fields, and localized DEP forces guide them to the designated receptor sites. This synergy enables the controlled, high-precision placement of micro-LEDs.

Micro-LEDs are directed toward the receptor holes by magnetic forces produced by a rotating magnetic rod array located beneath the substrate. The nickel embedded within the micro-LEDs facilitated movement in the fluid medium, and the rotation of the rods ensured an even distribution (Figure 8(e)). As the micro-LEDs approach the receptor holes, the DEP forces induced by the assembly electrodes capture and align them accurately. Additionally, titanium coatings enhance the DEP effect, and shape-matched receptor holes ensure the proper placement of RGB micro-LEDs. Each LED type (red, green, and blue) was designed with unique dimensions to optimize alignment and performance.

The MDSAT process begins with laser LLO, which releases micro-LEDs embedded in a ferromagnetic nickel layer from a sapphire donor substrate. After the release, the micro-LEDs are suspended in a water-based solution and directed toward the assembly substrate. In this process, the magnetic forces provide coarse positioning, whereas the DEP forces achieve fine alignment. The embedded ferromagnetic nickel layer facilitates magnetic control, guiding the micro-LEDs toward general target areas under an external magnetic field. Because the micro-LEDs are near their receptor sites, the localized DEP forces acting on the titanium layer further refine their positioning, aligning them precisely within the receptor holes.

Magnetic forces provide consistent long-range guidance, directing the micro-LEDs toward the receptor holes. When the micro-LEDs approach the receptor sites, the DEP force becomes dominant owing to the localized electric field gradient generated by the assembly electrodes. This sharp increase in DEP force near the receptor boundaries ensures precise alignment and secure trapping of the micro-LEDs. The dielectric layer amplifies the DEP effect, enabling accurate placement, while the magnetic forces facilitate large-scale transport across the substrate with high assembly efficiency (Figure 8(f)). Modifying the receptor hole height proved pivotal for optimizing the micro-LED assembly process because it directly enhanced the alignment accuracy and minimized defect rates. Increasing the receptor hole height from 3.0 μm to 4.5 μm markedly reduced shape-mismatch defects by facilitating the removal of misaligned micro-LEDs

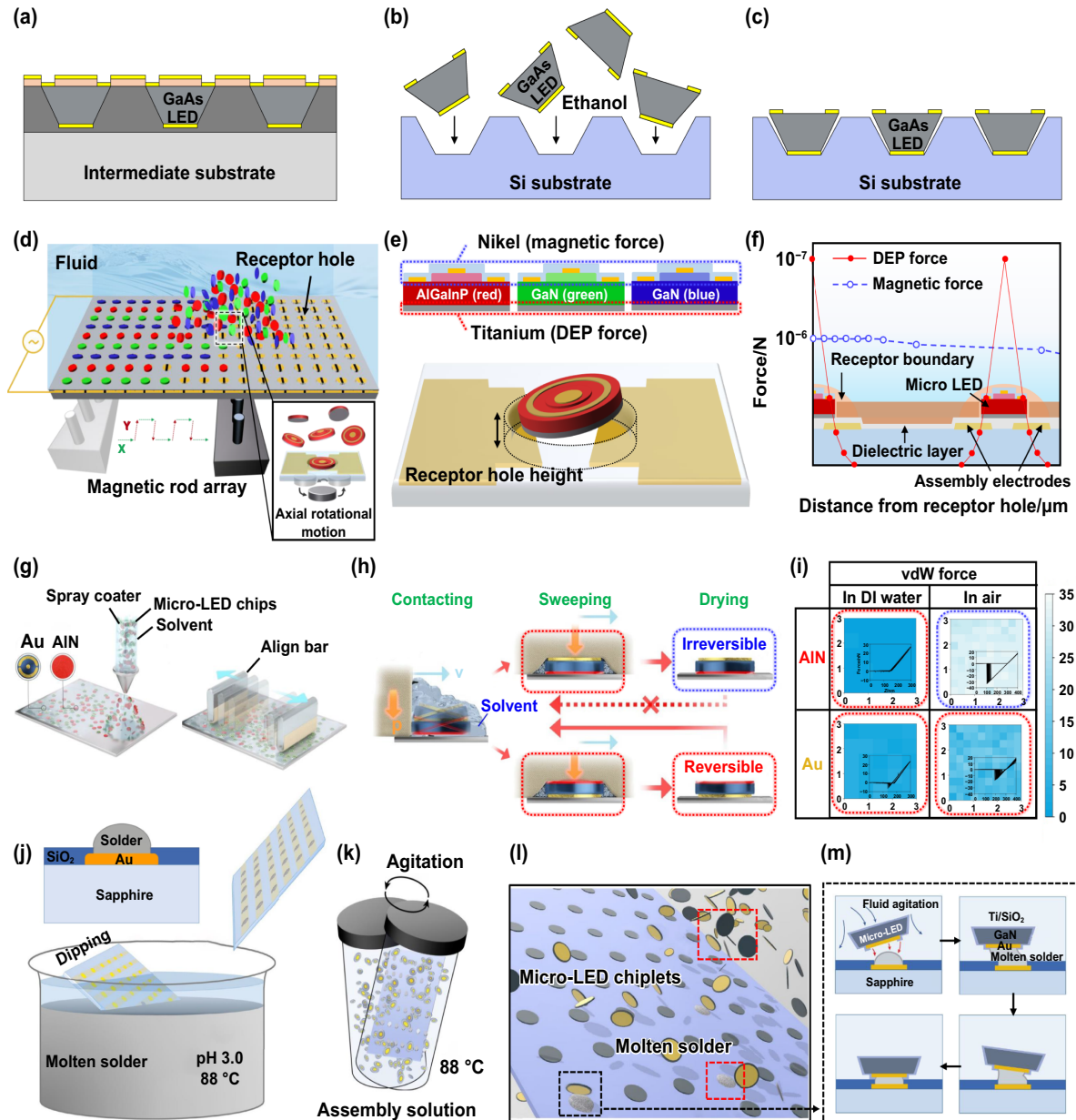


Figure 8. Fluidic self-assembly (a) Preparation of trapezoidal GaAs LED blocks on an intermediate substrate. (b) Fluidic self-assembly process in ethanol. (c) Final integrated state after fluid evaporation. (d) Schematic illustration of magnetic-force-assisted dielectrophoretic self-assembly (MDSAT) for RGB micro-LEDs. (e) Structure of RGB micro-LEDs integrated with material-specific layers to respond to external forces: nickel enables magnetic responsiveness, and titanium enhances the DEP effect. (f) Comparison of DEP and magnetic forces acting on micro-LEDs relative to distance from the receptor boundary. Adapted from^[163], with permission from Springer Nature. (g) Concept of wafer-scale micro-LED alignment using engineered vdW interactions. (h) Mechanism of engineered vdW interactions. (i) Quantitative mapping of vdW forces for AlN and Au surfaces under both wet and dry conditions. Adapted from^[239], with permission from Springer Nature. (j) Substrate preparation and dip soldering process. (k) Micro-LED chiplets dispersed in a heated aqueous assembly solution. (l) Schematic of micro-LED chiplets being guided toward molten solder bumps by fluidic motion. (m) Stepwise illustration of the solder-assisted self-assembly process. Adapted from^[240], with permission from Springer Nature.

during the magnetic force-guided assembly. This adjustment improved the coordination between the DEP and magnetic forces, ensuring that only properly aligned micro-LEDs were securely captured at the receptor sites, resulting in a transfer yield of 99.99%.

This highly efficient assembly method integrates magnetic and DEP forces, and is scalable to large substrates. This approach relies on precise receptor design and material

integration, which may constrain the cost efficiency in mass production; however, it represents a significant advancement in next-generation display manufacturing.

FSA combined with engineered vdW forces offers a highly efficient approach for the wafer-scale alignment and integration of micro-LEDs (Figure 8(g))^[239]. By designing two faces of micro-LEDs to exhibit distinct vdW interactions—achieved through tailored surface roughness and material

selection—this method enables single-face preferential alignment and selective bonding to the substrate.

Micro-LEDs feature engineered surfaces: a roughened Au face yielding weak vdW forces and an atomically smooth AlN face ensuring strong adhesion. Anisotropic etching with KOH solution removed the sacrificial Si layer and separated the micro-LEDs from the Si substrate without harming the AlN surface. The chips are dispersed in a solvent and fed into a fluidic self-alignment system. An alignment bar applies controlled pressure in the “contacting” phase to reorient the chips, while lateral movement in the “sweeping” stage positions them into target sites. In the “drying” phase, restored vdW forces irreversibly bond the AlN face to the substrate, with misaligned chips—owing to the weaker adhesion of the Au face—being corrected (Figure 8(h)).

A significant difference in the vdW forces between the two faces was critical (Figure 8(i)). Under dry conditions, the smooth AlN face exhibited a stronger vdW force (~ 30.7 nN) than the rough Au face (~ 11.8 nN), ensuring irreversible bonding when the AlN face contacted the substrate. Under wet conditions, the forces drop substantially (~ 0.36 nN for AlN and ~ 3.66 nN for Au), allowing the micro-LEDs to reorient freely. This disparity is further exploited during alignment correction, in which chips with the Au face down detach easily and are reoriented to converge on the desired AlN-down configuration, resulting in high alignment accuracy. This study demonstrates a highly efficient micro-LED assembly method that leverages engineered vdW forces to achieve precise single-face alignment with 99.992% yield. The process is scalable for large-area substrates and is well-suited for next-generation display manufacturing. This method relies on precise surface engineering and solvent drying processes, which may complicate mass production, whereas contamination in receptor holes can limit the transfer yield, requiring meticulous substrate preparation. This approach represents a significant advancement and offers a scalable, high-precision alternative to the traditional pick-and-place techniques for micro-LED integration.

By precisely controlling the solution viscosity, the FSA technique enables high-throughput micro-LED transfer (Figure 8(j))^[240]. Incorporating poloxamer—a viscosity-enhancing polymer—boosts liquid-to-chiplet momentum transfer and overcomes the low inertia challenges of sub-100 μm chiplets. This approach supports rapid large-scale assembly with exceptional alignment accuracy and yield.

The process begins with fabricating the substrate by depositing multiple layers of Cr, Pt, and Au onto a sapphire wafer, followed by patterning to form gold electrodes. A SiO_2 layer is then deposited and etched to expose the Au regions. Molten solder bumps are subsequently formed on these regions by dip soldering in a heated acidic solution (88°C , pH 3.0), which prevents oxidation (Figure 8(k)). The micro-LED chips are prepared and dispersed in an assembly solution comprising water, 2 wt% poloxamer, and acetic acid. When the substrate and chiplets are combined in this solution, controlled agitation guides the chiplets toward the molten solder bumps (Figure 8(l)). Upon contact, the micro-LEDs align and bond

with the solder bumps, and as the solder cools and solidifies, the chiplets are securely anchored in their designated positions (Figure 8(m)).

The optimization of the assembly process involved fine-tuning the poloxamer concentration. At 2 wt%, the solution's viscosity increased by 55% compared to pure water (0.50 mPa·s vs. 0.32 mPa·s at 88°C), enhancing momentum transfer and yielding over 97%. In contrast, a 4 wt% concentration resulted in excessive viscosity (0.85 mPa·s), reducing chiplet velocity and lowering the assembly yield ($\sim 15\%$) due to increased viscous dissipation. This optimization identifies an ideal viscosity range (~ 0.49 mPa·s) that balances fluid velocity with chiplet momentum to maximize alignment precision and yield.

Under controlled viscosity conditions and optimized chiplet design, a 2 wt% poloxamer solution enhances viscosity and facilitates precise alignment of $45\ \mu\text{m}$ micro-LED chiplets, resulting in a failure rate as low as 0.0012 (a 99.88% yield) over nearly 19 000 chiplets. Alignment error analysis reveals an average error of only $1.25\ \mu\text{m}$, with 99.5% of chiplets positioned within $2.9\ \mu\text{m}$. These results underscore the scalability and precision of this method and highlight its potential for cost-effective high-throughput micro-LED assembly in next-generation displays.

This efficient FSA technique optimized through viscosity control achieves a 99.88% yield and precise alignment for $45\ \mu\text{m}$ micro-LED chiplets. This method is scalable and cost-effective, making it suitable for high-throughput manufacturing. Its reliance on precise viscosity control and substrate preparation may complicate large-scale commercial implementation, and additional optimization is required to accommodate finer pitch dimensions for smaller displays and approach near-perfect yields. This technique represents a significant advancement in micro-LED manufacturing, offering a practical alternative to the traditional pick-and-place or transfer printing methods for next-generation display applications.

In fluidic self-assembly, the fluidic medium functions as both a transport medium for micro-LEDs and a stabilizer during drying. Ethanol and DI water are commonly used due to their low viscosity and compatibility with high-throughput alignment, but their surface tension can disturb chip alignment during evaporation. Supercritical CO_2 avoids this issue by eliminating surface tension, thereby preserving alignment accuracy. Viscosity is another key factor: adding poloxamer improves momentum transfer and alignment by tuning fluid resistance to chip size and inertia. Similarly, van der Waals-based single-face alignment benefits from controlled viscosity, which influences chip reorientation. Studies comparing solvents like IPA, DI water, glycerin, and honey reveal that while alignment remains effective, viscosity and surface tension shape the dynamics of reorientation. In magnetically and dielectrophoretically assisted methods, water-based media enable dominant field-driven control. Thus, fluidic media selection must balance viscosity, surface tension, chemical compatibility, and interaction with external forces to maximize transfer yield and scalability.

Table 3. Comparison of assembly techniques for micro-LEDs.

Technique	Key mechanism	Yield/%	Transfer rate	LED size	Damage	Selectivity	Print mode	Scalability	Cost
Elastomeric	Adhesive interaction	99.99%	7 million·h ⁻¹	>10 μm	Low	High	Contact	Moderate	Low
Roll-to-roll	Adhesive interaction	99.99%	144 million·h ⁻¹	<100 μm	Low	Low	Contact	High	Low
Electrostatic	Electrostatic force	N/A	12 million·h ⁻¹	1–100 μm	High	Moderate	Contact	Low	High
Electromagnetic	Electromagnetic force	N/A	0.9 million·h ⁻¹	>10 μm	High	High	Contact	Low	High
Laser	Heating & Ablation	99.99%	500 million·h ⁻¹	>1 μm	High	High	Non-contact	High	High
Microvacuum	Suction force	98.06%	N/A	80 μm	Low	High	Contact	High	Low
Fluidic	Gravity & Capillary force	99.99%	100 million·h ⁻¹	5–200 μm	High	Low	Non-contact	Moderate	Moderate

8. Conclusions and outlook

In this review, we summarize the latest advancements in assembly techniques for micro-LEDs, which are characterized by unique mechanisms that enhance transfer efficiency and scalability (Table 3). Techniques, such as elastomeric, roll-to-roll, electrostatic, electromagnetic, laser, microvacuum, and fluidic methods, have distinct advantages and limitations. A common achievement of these methods is high yield, often reaching 99.99%, which marks significant progress in microassembly technology. However, even this yield is insufficient for mass-producing ultrahigh-resolution displays, where minor defects can significantly degrade the quality. For example, an 8 K UHD display with 7680×4320 pixels requires approximately 100 million RGB micro-LEDs. A 99.99% yield leaves approximately 10 000 defective pixels, leading to costly inspection and repairs, ultimately increasing production costs and limiting scalability. To achieve true commercial viability, even higher yields (approaching 99.999 99%) are required.

Transfer yield is influenced by a variety of interdependent factors, including adhesion control, chip damage, placement accuracy, and bonding reliability, among others. Adhesion control plays a critical role during both the pick-up and printing stages, as the success of transfer depends on the ability to finely tune the adhesion contrast between the stamp–chip and chip–substrate interfaces. Chip damage can occur during the release or transfer process, especially in methods involving high-energy input, such as laser-based or electrostatic transfer, where thermal or electrical stress may degrade micro-LED performance. Placement accuracy becomes increasingly important as chip size decreases and pixel density increases; even slight misalignments can prevent proper electrical or optical contact. Bonding reliability is another decisive factor—chips that are correctly aligned may still fail to establish functional interconnections if the adhesive properties or bonding conditions are suboptimal. Beyond these, a range of other factors—including process uniformity, environmental stability, and surface planarity—also contribute to overall yield. Importantly, these factors do not act independently; rather, they are intricately coupled and often involve trade-offs. For instance, increasing adhesion strength may improve pick-up reliability but hinder clean release. As such, the relationship among these factors remains complex and not yet fully optimized, underscoring the need for holistic approaches to yield engineering.

For large-area processes, adhesives—especially anisotropic conductive adhesives (ACA)—are essential to ensure mechanical stability and electrical connectivity^[181,241,242]. The integration of ACA has been shown to significantly improve yield in large-area transfers by enabling simultaneous bonding and alignment. However, yield is not determined by any single factor; instead, it depends on the complex interaction among multiple parameters. For instance, while the pedestal structure improves pick-up efficiency, it can severely hinder the printing step, ultimately lowering overall yield. Thus, optimizing these key factors in combination is crucial to enhancing the overall process.

Based on the strengths and weaknesses of each approach, current techniques have been selectively applied to different display formats. Table 3 provides a comparative overview of the major transfer techniques for micro-LED assembly, each characterized by distinct mechanisms, process parameters, and yields. Elastomeric stamp and roll-to-roll techniques both rely on adhesive interactions, achieving high yields (99.99%) and low damage, with R2R showing exceptional scalability and transfer speed, making these methods particularly suitable for low-to-mid PPI displays. Electrostatic and electromagnetic methods, utilizing force-based mechanisms, offer superior selectivity and are more suitable for small-pitch, high-resolution applications, though they face limitations in damage control and scalability. Laser-based transfer exhibits high yield and excellent selectivity in a non-contact configuration, positioning it as a strong candidate for precision-demanding applications. Microvacuum transfer, based on suction force, maintains low damage and high selectivity, offering a balance between precision and scalability. Fluidic self-assembly, though limited in selectivity, enables high-throughput non-contact transfer over a wide chip size range and is potentially useful for large-area displays or passive-matrix backplanes.

We believe that further improvements in transfer yield are not only possible but highly promising within current technological frameworks, provided that specific parameters unique to each method continue to be refined. For instance, in elastomeric stamp techniques, substantial advancements have already been made through structural innovations in the stamp itself—such as microstructured surfaces and pedestal geometries—that improve adhesion contrast. Continued development of novel stamp architectures is expected to further enhance adhesion control and release reliability. In the case of roll-to-roll systems, improvements in mechanical design and real-time alignment feedback have significantly

enhanced placement accuracy. As the precision of roll stamp systems and the understanding of web tension dynamics continue to advance, higher transfer yields can be achieved even for dense chip arrays.

For electrostatic and electromagnetic methods, yield can be further enhanced through detailed analysis of the electrical force-generation mechanisms. For example, the thickness of dielectric insulating layers, the geometry of pickup electrodes, and the mechanical configuration of the coils all influence force magnitude and distribution. Optimizing these parameters can lead to stronger, more selective pick-up capabilities while simultaneously minimizing electrical damage to the chips. In laser-based transfer, progress in the design of expandable layers or laser-active sacrificial films is expected to improve release uniformity. Tailoring the material composition and thickness of these layers may allow micro-LEDs to be released cleanly with minimal thermal stress, thus improving both accuracy and yield.

In microvacuum transfer, current demonstrations have achieved reliable handling of chips as small as 50 μm . However, by optimizing micro-hole size, vacuum channel geometry, and sealing conditions, the method could potentially be extended to handle much smaller devices, such as those in the 20 μm or even 10 μm range. Similarly, fluidic self-assembly holds further promise through enhanced control over the fluid medium. Investigations into chip-fluid interactions, as well as the effect of chip and receptor geometry on self-alignment behavior, can help improve placement accuracy and yield. Expanding the variety of fluid types and refining channel designs may also contribute to more deterministic assembly.

Taken together, these perspectives illustrate that while each technique has unique limitations, they also contain under-exploited opportunities for refinement. Just as past progress in adhesion control, alignment, or bonding has significantly advanced transfer yield, we anticipate that continued in-depth research within each methodological framework will unlock further yield improvements.

Beyond improving individual techniques, hybrid transfer strategies are emerging as a promising direction. For instance, shear-assisted alignment mechanisms can be combined with roll-to-roll systems to modulate adhesion through controlled shear strain, enabling sub-5 μm alignment accuracy during high-speed, large-area assembly^[207]. Similarly, pedestal-shaped stamp structures can be paired with laser-assisted printing, where the geometry-driven high pick-up stability is preserved while localized heating from the laser facilitates rapid release, addressing the inherent detachment limitations of pedestal designs^[198]. As another example, FSA combined with magnetic and DEP forces improves micro-LED control in liquid media, where magnetic forces guide the overall LED clusters toward the assembly region, and DEP forces drive final docking into receptor holes for precise alignment and positioning^[163]. In another scenario, micro-structured elastomeric stamps could be integrated with electrostatic or electromagnetic forces for active alignment and selective release. These examples highlight that no transfer method needs to operate in isolation—hybrid strategies that leverage complementary strengths offer significant potential to

overcome current barriers. To maximize the benefits of hybrid strategies, the combined techniques must be mutually compatible, and potential trade-offs should be carefully managed to ensure that the integration of strengths does not introduce new weaknesses, ultimately driving improvements in transfer yield.

In parallel, researchers are working to optimize substrate compatibility, adhesion control, and overall transfer mechanisms to meet the needs of specific applications. Particularly in advanced systems like near-eye displays for AR glasses, technologies such as beam shaping, optical coupling, and pixel-level light control are essential^[243–246]. Improvements in transfer yield must be accompanied by compatibility with these optical modules. Customized assembly processes tailored to the unique constraints of each device type—whether flexible, wearable, or high-resolution—will be necessary to support practical deployment across the display industry.

Another important consideration for practical deployment is the development of effective repair strategies. Even with extremely high yields, some defects are inevitable. Replacing or repairing defective pixels through localized re-transfer or selective correction processes can prevent entire panels from being discarded, reducing waste and production cost. Therefore, along with improving first-pass yield, robust repair technologies are essential for the industrial viability of micro-LED displays.

The most critical bottleneck for the industrialization of micro-LEDs is the excessive cost that arises from yield loss. As chip sizes shrink and pixel densities increase, even minor defects can lead to repair processes that are both technically challenging and prohibitively expensive. Thus, achieving near-perfect transfer yield is the single most important requirement for reducing cost and enabling commercialization. This review emphasizes transfer processes and concludes that two technical strategies are essential for maximizing yield: precise control of key process parameters—such as adhesion, damage, and alignment—and continuous innovation through advanced or hybrid transfer mechanisms.

Beyond this central imperative, additional strategies are also being actively explored to alleviate cost pressures. One is to balance yield with repair: in large-area displays where repair can be implemented more readily, it may be more cost-effective to accept a modest defect rate and use repair to achieve a “just-enough” specification, rather than striving for absolute perfection. Another is process integration, where transfer, bonding, and even repair are combined into a unified workflow^[247]. For example, laser-based methods that simultaneously release chips, activate adhesives, and reroute defective pixels illustrate how such integration can shorten cycle times and reduce costs^[248]. In parallel, close collaboration among equipment makers, process developers, and display manufacturers will be critical to defining realistic tool requirements that deliver sufficient performance without unnecessary over-specification, thereby ensuring cost-optimized manufacturing. Collectively, these approaches reflect the multiple pathways through which the industry is working toward the ultimate goal of achieving the commercial viability of

micro-LEDs, with yield enhancement remaining the central priority.

Given the rapidly growing interest and investment in this field, the commercialization of micro-LED technology appears increasingly attainable. However, transformative breakthroughs are still possible, as current methods have not yet reached their full potential. The broad diversity of display configurations further demands that transfer technologies be flexible, adaptable, and scalable. Continuous innovation—whether through hybrid methods, process integration, or advanced repair strategies—will be essential to achieving the ultra-high yields necessary for commercialization.

In summary, although significant technical hurdles remain, the pathway toward 99.999 99% yield is being actively pursued. Through interdisciplinary collaboration and sustained innovation, micro-LED assembly is approaching the level of maturity required for large-scale manufacturing. These collective efforts point toward a future in which ultra-high-resolution, defect-free micro-LED displays become a commercial reality.

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References

- [1] Chang N, Choi I and Shim H. 2004. DLS: dynamic backlight luminance scaling of liquid crystal display. *IEEE Trans. Very Large Scale Integr. VLSI Syst.* **12**, 837–846.
- [2] Chen H W, Tan G J and Wu S-T. 2017. Ambient contrast ratio of LCDs and OLED displays. *Opt. Express*. **25**, 33643–33656.
- [3] Chen H-W, Lee J-H, Lin B-Y, Chen S and Wu S-T. 2018. Liquid crystal display and organic light-emitting diode display: present status and future perspectives. *Light Sci. Appl.* **7**, 17168.
- [4] Geffroy B, Le Roy P and Prat C. 2006. Organic light-emitting diode (OLED) technology: materials, devices and display technologies. *Polym. Int.* **55**, 572–582.
- [5] Henry W and Percival C. 2016. 55-2: invited Paper: ILED displays: next generation display technology. In *SID Symp. Digest of Technical Papers* vol 47, pp 747–750.
- [6] Jia H P. 2018. Who will win the future of display technologies? *Natl Sci. Rev.* **5**, 427–431.
- [7] Schadt M. 2009. Milestone in the history of field-effect liquid crystal displays and materials. *Jpn. J. Appl. Phys.* **48**, 03B001.
- [8] Tan G J, Huang Y G, Li M-C, Lee S-L and Wu S-T. 2018. High dynamic range liquid crystal displays with a mini-LED backlight. *Opt. Express* **26**, 16572–16584.
- [9] Tang C W and VanSlyke S A. 1987. Organic electroluminescent diodes. *Appl. Phys. Lett.* **51**, 913–915.
- [10] Weber L F. 2006. History of the plasma display panel. *IEEE Trans. Plasma Sci.* **34**, 268–278.
- [11] Yoon J K, Park E M, Son J S, Shin H W, Kim H E, Yee M, Kim H G, Oh C H and Ahn B C. 2013. 27.2: the study of picture quality of OLED TV with WRGB OLEDs structure. In *SID Symp. Digest of Technical Papers* vol 44, pp 326–329.
- [12] Jiang H X and Lin J Y. 2013. Nitride micro-LEDs and beyond—a decade progress review. *Opt. Express* **21**, A475–A484.
- [13] Wierer J J Jr and Tansu N. 2019. III-nitride micro-LEDs for efficient emissive displays. *Laser Photonics Rev.* **13**, 1900141.
- [14] Wong M S, Nakamura S and DenBaars S P. 2020. Review—progress in high performance III-nitride micro-light-emitting diodes. *ECS J. Solid State Sci. Technol.* **9**, 015012.
- [15] Bandari V K and Schmidt O G. 2024. A bright future for micro-LED displays. *Light Sci. Appl.* **13**, 317.
- [16] Kim D-H, Lu N S, Ghaffari R and Rogers J A. 2012. Inorganic semiconductor nanomaterials for flexible and stretchable bio-integrated electronics. *npg Asia Mater.* **4**, e15.
- [17] Kim R-H et al. 2012. Materials and designs for wirelessly powered implantable light-emitting systems. *Small* **8**, 2812–2818.
- [18] Lee H E et al. 2018. Monolithic flexible vertical GaN light-emitting diodes for a transparent wireless brain optical stimulator. *Adv. Mater.* **30**, 1800649.
- [19] Lee H E, Lee S H, Jeong M, Shin J H, Ahn Y, Kim D, Oh S H, Yun S H and Lee K J. 2018. Trichogenic photostimulation using monolithic flexible vertical AlGaInP light-emitting diodes. *ACS Nano*. **12**, 9587–9595.
- [20] Lee H E, Shin J H, Park J H, Hong S K, Park S H, Lee S H, Lee J H, Kang I-S and Lee K J. 2019. Micro light-emitting diodes for display and flexible biomedical applications. *Adv. Funct. Mater.* **29**, 1808075.
- [21] Lee S H, Kim J, Shin J H, Lee H E, Kang I-S, Gwak K, Kim D-S, Kim D and Lee K J. 2018. Optogenetic control of body movements via flexible vertical light-emitting diodes on brain surface. *Nano Energy* **44**, 447–455.
- [22] Lee S Y, Park K-I, Huh C, Koo M, Yoo H G, Kim S, Ah C S, Sung G Y and Lee K J. 2012. Water-resistant flexible GaN LED on a liquid crystal polymer substrate for implantable biomedical applications. *Nano Energy* **1**, 145–151.
- [23] Moore S K. 2018. The nextgen display: microleds [News]. *IEEE Spectr.* **55**, 9–10.
- [24] Park G et al. 2014. Immunologic and tissue biocompatibility of flexible/stretchable electronics and optoelectronics. *Adv. Healthcare Mater.* **3**, 515–525.

- [25] Park J et al. 2021. Design and fabrication of blue LED-integrated graphene electrodes for neural stimulation and signal recording. *ACS Appl. Electron. Mater.* **3**, 4308–4316.
- [26] Park S I et al. 2015. Soft, stretchable, fully implantable miniaturized optoelectronic systems for wireless optogenetics. *Nat. Biotechnol.* **33**, 1280–1286.
- [27] Seo S J, Park S and Jang H W. 2025. Flexible micro-LEDs: advanced fabrication techniques and applications. *Electron. Mater. Lett.* **21**, 311–330.
- [28] Wu T Z, Sher C-W, Lin Y, Lee C-F, Liang S J, Lu Y J, Huang Chen S-W, Guo W J, Kuo H-C and Chen Z. 2018. Mini-LED and micro-LED: promising candidates for the next generation display technology. *Appl. Sci.* **8**, 1557.
- [29] Zhang H and Rogers J A. 2019. Recent advances in flexible inorganic light emitting diodes: from materials design to integrated optoelectronic platforms. *Adv. Opt. Mater.* **7**, 1800936.
- [30] Zhou X J, Tian P F, Sher C-W, Wu J, Liu H Z, Liu R and Kuo H-C. 2020. Growth, transfer printing and colour conversion techniques towards full-colour micro-LED display. *Prog. Quantum Electron.* **71**, 100263.
- [31] Kim R-H et al. 2010. Waterproof AlInGaP optoelectronics on stretchable substrates with applications in biomedicine and robotics. *Nat. Mater.* **9**, 929–937.
- [32] Ahmed K 2019. Heterogeneous micro LED displays yield statistics. In *2019 IEEE SOI-3D-Subthreshold Microelectronics Technology Unified Conference (S3S)* (IEEE, San Jose, CA, USA) 1–2.
- [33] Bower C A et al 2021. 60-1: invited Paper: mass transfer throughput and yield using elastomer stamps. In *SID Symp. Digest of Technical Papers* vol 52, pp 849–852.
- [34] Ding K, Avrutin V, Izyumskaya N, Özgür Ü and Morkoç H. 2019. Micro-LEDs, a manufacturability perspective. *Appl. Sci.* **9**, 1206.
- [35] Gomez D, Ghosal K, Moore T, Meitl M A, Bonafede S, Prevatte C, Radauscher E, Trindade A J and Bower C A. 2017. Scalability and yield in elastomer stamp micro-transfer-printing. In *2017 IEEE 67th Electronic Components and Technology Conference (ECTC)* (IEEE, Orlando, FL, USA) pp 1779–1785.
- [36] Henley F J 2018. 9.1: invited Paper: materials, process and production equipment considerations to achieve high-yield MicroLED mass-production. In *SID Symp. Digest of Technical Papers*. vol 49, pp 86–89.
- [37] Lee V W, Twu N and Kyymissis I. 2016. Micro-LED technologies and applications. *Inform. Disp.* **32**, 16–23.
- [38] Lin -C-C et al. 2023. The micro-LED roadmap: status quo and prospects. *J. Phys. Photonics* **5**, 042502.
- [39] Linghu C H, Zhang S, Wang C J, Luo H Y and Song J Z. 2021. Mass transfer for Micro-LED display: transfer printing techniques. *Semiconduct. Semimet.* **106**, 253–280.
- [40] Ogier S and Hendy I. 2025. Partnership leads to new generation of rollable OTFT MicroLED displays. *Inform. Disp.* **41**, 19–24.
- [41] Virey E H, Baron N and Bouhamri Z 2019. 11-3: overlooked challenges for microLED displays. In *SID Symp. Digest of Technical Papers*. vol 50, pp 129–132.
- [42] Yu B H, Li Y, Li J S, Ding X R and Li Z T. 2024. Challenges of high-yield manufacture in micro-light-emitting diodes displays: chip fabrication, mass transfer, and detection. *J. Appl. Phys.* **57**, 463001.
- [43] Zhu G Q, Liu Y J, Ming R, Shi F and Cheng M J. 2022. Mass transfer, detection and repair technologies in micro-LED displays. *Sci. China Mater.* **65**, 2128–2153.
- [44] Bourget C M. 2008. An introduction to light-emitting diodes. *HortScience* **43**, 1944–1946.
- [45] Burroughes J H, Bradley D D C, Brown A R, Marks R N, Mackay K, Friend R H, Burns P L and Holmes A B. 1990. Light-emitting diodes based on conjugated polymers. *Nature* **347**, 539–541.
- [46] Chang M-H, Das D, Varde P V and Pecht M. 2012. Light emitting diodes reliability review. *Microelectron. Reliab.* **52**, 762–782.
- [47] Round H J. 1991. A note on carborundum. *Semiconductor Devices: Pioneering Papers* (World Scientific) p 879.
- [48] Schubert E F 2018. *Light-emitting diodes*. 3rd edn https://books.google.co.kr/books?hl=ko&lr=&id=GEFKDwAAQBAJ&oi=fnd&pg=PP1&ots=t7fpYWqlQV&sig=I24j3rlvtqjmhj0G_4-fmiQ9DE&redir_esc=y#v=onepage&q&f=false.
- [49] Ahn J H and Kim J H. 2021. *Micro Light Emitting Diode: Fabrication and Devices*. (Springer, Singapore).
- [50] Amano H, Kito M, Hiramatsu K and Akasaki I. 1989. P-type conduction in Mg-doped GaN treated with low-energy electron beam irradiation (LEEBI). *Jpn. J. Appl. Phys.* **28**, L2112–L2114.
- [51] Amano H, Sawaki N, Akasaki I and Toyoda Y. 1986. Metalorganic vapor phase epitaxial growth of a high quality GaN film using an AlN buffer layer. *Appl. Phys. Lett.* **48**, 353–355.
- [52] Behrman K and Kyymissis I. 2022. Micro light-emitting diodes. *Nat. Electron.* **5**, 564–573.
- [53] Holonyak J N and Bevacqua S F. 1962. Coherent (visible) light emission from Ga(As_{1-x}P_x) junctions. *Appl. Phys. Lett.* **1**, 82–83.
- [54] Jiang H X and Lin J Y. 2021. *Micro LEDs*. (Academic Press, Amsterdam).
- [55] Jin S X, Li J, Li J Z, Lin J Y and Jiang H X. 2000. GaN microdisk light emitting diodes. *Appl. Phys. Lett.* **76**, 631–633.
- [56] Jin S X, Li J, Lin J Y and Jiang H X. 2000. InGaN/GaN quantum well interconnected microdisk light emitting diodes. *Appl. Phys. Lett.* **77**, 3236–3238.
- [57] Lee D S and Han J H. 2021. Micro-LED technology for display applications. In *Advanced Display Technology: Next Generation Self-Emitting Displays* (eds Kang I B, Han C W, Jeong J K) (Springer, Singapore) pp 271–305.
- [58] Li Y L and Liu Y T. 2020. MicroLED display: the next-generation display technology. *Proc. SPIE* **11304**, 113040H.
- [59] Nakamura S, Mukai T, Senoh M S M and Iwasa N I N. 1992. Thermal annealing effects on P-type Mg-doped GaN films. *Jpn. J. Appl. Phys.* **31**, L139–L142.
- [60] Nakamura S, Senoh M, Iwasa N and Nagahama S I N S I. 1995. High-brightness InGaN blue, green and yellow light-emitting diodes with quantum well structure. *Jpn. J. Appl. Phys.* **34**, L797.
- [61] Parbrook P J, Corbett B, Han J N, Seong T-Y and Amano H. 2021. Micro-light emitting diode: from chips to applications. *Laser Photonics Rev.* **15**, 2000133.
- [62] Zhang S H et al. 2023. Research progress of micro-LED display technology. *Crystals* **13**, 1001.
- [63] Chang S H, Koo J H, Yoo J, Kim M S, Choi M K, Kim D-H and Song Y M. 2024. Flexible and stretchable light-emitting diodes and photodetectors for human-centric optoelectronics. *Chem. Rev.* **124**, 768–859.
- [64] Gao Z W, Ning H L, Yao R H, Xu W, Zou W X, Guo C X, Luo D X, Xu H R and Xiao J L. 2022. Mini-LED backlight technology progress for liquid crystal display. *Crystals* **12**, 313.
- [65] Hu X L, Krull P, de Graff B, Dowling K, Rogers J A and Arora W J. 2011. Stretchable inorganic-semiconductor electronic systems. *Adv. Mater.* **23**, 2933–2936.
- [66] Huang Y G, Tan G J, Gou F W, Li M-C, Lee S-L and Wu S-T. 2019. Prospects and challenges of mini-LED and micro-LED displays. *J. Soc. Inform. Disp.* **27**, 387–401.
- [67] Kim N, Kim J, Seo J, Hong C and Lee J. 2022. Stretchable inorganic LED displays with double-layer modular design

- for high fill factor. *ACS Appl. Mater. Interfaces* **14**, 4344–4351.
- [68] Lei J H, Zhu H L, Huang X, Lin J J, Zheng Y F, Lu Y J, Chen Z and Guo W J. 2024. Mini-LED backlight: advances and future perspectives. *Crystals* **14**, 922.
- [69] Miyakawa M, Tsuji H, Takei T, Yamamoto T, Fujisaki Y and Nakata M. 2025. Highly stretchable LED display using liquid metal and molybdenum-barriered multilayer electrodes with long-term reliability. *Adv. Electron. Mater.* **11**, 2400676.
- [70] Chong W C, Cho W K, Liu Z J, Wang C H and Lau K M. 2014. 1700 pixels per inch (PPI) passive-matrix micro-LED display powered by ASIC In *2014 IEEE Compound Semiconductor Integrated Circuit Symposium (CSICS)* (IEEE, La Jolla, CA, USA) pp 1–4.
- [71] Martin P S, Screenivasan R and Sherwood T. 2024. Micro-LED for AR applications. *Proc. SPIE* **12906**, 1290602.
- [72] Qi L H, Zhang X, Chong W C and Lau K M. 2023. Monolithically integrated high-resolution full-color GaN-on-Si micro-LED microdisplay. *Photon. Res.* **11**, 117.
- [73] Qi L H, Zhang X, Chong W C, Li P A and Lau K M. 2021. 848 ppi high-brightness active-matrix micro-LED microdisplay using GaN-on-Si epi-wafers towards mass production. *Opt. Express* **29**, 10580–10591.
- [74] Zhang X, Li P A, Zou X B, Jiang J M, Yuen S H, Tang C W and Lau K M. 2019. Active matrix monolithic LED micro-display using GaN-on-Si epilayers. *IEEE Photonics Technol. Lett.* **31**, 865–868.
- [75] Chu C-F, Lai F-I, Chu J-T, Yu C-C, Lin C-F, Kuo H-C and Wang S. 2004. Study of GaN light-emitting diodes fabricated by laser lift-off technique. *J. Appl. Phys.* **95**, 3916–3922.
- [76] Kim J, Kim J-H, Cho S-H and Whang K-H. 2016. Selective lift-off of GaN light-emitting diode from a sapphire substrate using 266-nm diode-pumped solid-state laser irradiation. *Appl. Phys. A* **122**, 305.
- [77] Kim T-I et al. 2012. High-efficiency, microscale GaN light-emitting diodes and their thermal properties on unusual substrates. *Small* **8**, 1643–1649.
- [78] Lee S H, Park S Y and Lee K J. 2012. Laser lift-off of GaN thin film and its application to the flexible light emitting diodes. *Proc. SPIE* **8460**, 846011.
- [79] Li S-H, Lin C-P, Fang Y-H, Kuo W-H, Wu M-H, Chao C-L, Horng R-H and Su G-D J. 2019. Performance analysis of GaN-based micro light-emitting diodes by laser lift-off process. *Solid State Electron. Lett.* **1**, 58–63.
- [80] Lin C C, Fang Y H, Kao M J, Huang P K, Chang F P, Yang L C and Wu C I. 2018. 59-2: invited Paper: ultra-fine pitch thin-film micro LED display for indoor applications In *SID Symp. Digest of Technical Papers*. vol 49, pp 782–785.
- [81] Tian W Y, Wu Y S, Xiao J Q, Wang P K and Li J H. 2023. Laser lift-off mechanism and optical-electric characteristics of red Micro-LED devices. *Opt. Express* **31**, 7887–7899.
- [82] Ueda T, Ishida M and Yuri M. 2011. Separation of thin GaN from sapphire by laser lift-off technique. *Jpn. J. Appl. Phys.* **50**, 041001.
- [83] Vögl F, Avramescu A, Gelfert S, Lex A, Waag A, Hetzl M and von Malm N. 2024. Optical characteristics of thin film-based InGaP micro-LED arrays: a study on size effect and far field behavior. *Opt. Express* **32**, 17644–17656.
- [84] Wu Y S, Cheng J-H, Peng W C and Ouyang H. 2007. Effects of laser sources on the reverse-bias leakages of laser lift-off GaN-based light-emitting diodes. *Appl. Phys. Lett.* **90**, 251110.
- [85] Yulianto N, Kadja G T M, Bornemann S, Gahlawat S, Majid N, Triyana K, Abdi F F, Wasisto H S and Waag A. 2021. Ultrashort pulse laser lift-off processing of InGaP/GaN light-emitting diode chips. *ACS Appl. Electron. Mater.* **3**, 778–788.
- [86] Wong W S, Sands T, Cheung N W, Kneissl M, Bour D P, Mei P, Romano L T and Johnson N M. 1999. Fabrication of thin-film InGaP light-emitting diode membranes by laser lift-off. *Appl. Phys. Lett.* **75**, 1360–1362.
- [87] LEDinside. 2020. Latest mini LED and micro LED definition and technology analysis. https://www.ledinside.com/news/2020/7/miniled_microled_comparison.
- [88] Lau J H et al. 2021. Fan-out panel-level packaging of mini-LED RGB display. *IEEE Trans. Compon. Packag. Manuf. Technol.* **11**, 739–747.
- [89] Paranjpe A, Montgomery J, Lee S M and Morath C. 2018. 45-2: invited Paper: micro-LED displays: key manufacturing challenges and solutions. In *SID Symp. Digest of Technical Papers* vol 49, pp 597–600.
- [90] Phenix S J. 2017. 43-1: invited Paper: display technology trends in digital signage displays. In *SID Symp. Digest of Technical Papers* vol 48, pp 608–611.
- [91] Cho J and Kim J K. 2019. Transfer or delivery of micro light-emitting diodes for light-emitting diode displays. *AIP Adv.* **9**, 100901.
- [92] Kang C S, Kim M, Kim S, Ahn H, Lee K, Han S and Park C J. 2021. Micro led display with printed circuit board assembly. *U.S. Patent*. No. 20210167055A1.
- [93] Lin J Y and Jiang H X. 2020. Development of microLED. *Appl. Phys. Lett.* **116**, 100502.
- [94] Park C-W, Lee M-E, Lee S-T and Song Y-W. 2020. Analysis of technological evolution for TV displays using S-Curves. *APJCRJ*. **6**, 147–163.
- [95] Hendy I and Gray P. 2020. The role of innovation in the display industry's end game. *Inform. Disp.* **36**, 28–32.
- [96] Hsiang E-L, He Z Q, Huang Y G, Gou F W, Lan Y-F and Wu S-T. 2020. Improving the power efficiency of micro-LED displays with optimized LED chip sizes. *Crystals* **10**, 494.
- [97] Mertens R. 2024. MicroLED displays: industry status and roadmap. *Inform. Disp.* **40**, 5–8.
- [98] Robin I C, Tchoulfian P, Sanniccolo T, Malier Y, Butaud E and Gilet P. 2024. 39-1: invited Paper: RGB nanowire based micro-LED chips for efficient and high performance ultra fine pitch direct view displays. In *SID Symp. Digest of Technical Papers* vol 55, pp 334–338.
- [99] Chen F and Kuo J. 2022. In-vehicle display technology. In *Advanced Driver Assistance Systems and Autonomous Vehicles: From Fundamentals to Applications* (eds Li Y, Shi H L) (Springer, Singapore) pp 339–419.
- [100] Yang C L, Lin C L, Hsieh H S, Kuo S M, Weng R, Kuo K H, Hsieh T H, Hsieh C Y, Chang S H and Wu Y H. 2024. 11-4: invited Paper: advanced augmented reality head-up display utilizing micro LED display technology. In *SID Symp. Digest of Technical Papers*. vol 55, pp 115–118.
- [101] Zhou C, Qiao W, Hua J Y and Chen L S. 2024. Automotive augmented reality head-up displays. *Micromachines* **15**, 442.
- [102] Ahmed K. 2024. *MicroLED Devices and Systems*. (Artech, Norwood).
- [103] YOLE Group. 2024. Did Apple just kill the microLED industry?. <https://www.yolegroup.com/strategy-insights/did-apple-just-kill-the-microled-industry/>.
- [104] Jeon D-H, Jeong W-B, Chung H-J and Lee S-W. 2022. Novel micro-LED display featuring fingerprint recognition without additional sensors. *IEEE Access* **10**, 74187–74197.
- [105] Kale E. 2023. The future is MicroLEDs, given the long gestation of display technologies. <https://displaydaily.com/the-future-is-microleds-given-the-long-gestation-of-display-technologies/>.
- [106] Kang C-M and Lee H. 2022. Recent progress of organic light-emitting diode microdisplays for augmented reality/virtual reality applications. *J. Inform. Disp.* **23**, 19–32.

- [107] Kashyap K, Galanis E and Zhu T T. 2023. Simplifying the manufacture and design of optical systems for AR glasses with DynamicPixelTuning® microLED technology. *Proc. SPIE* **12449**, 124490M.
- [108] Liu Z J, Chong W C, Wong K M and Lau K M. 2015. GaN-based LED micro-displays for wearable applications. *Microelectron. Eng.* **148**, 98–103.
- [109] Liu Z J, Chong W C, Wong K A and Lau K M. 2013. 360 PPI flip-chip mounted active matrix addressable light emitting diode on silicon (LEDOS) micro-displays. *J. Disp. Technol.* **9**, 678–682.
- [110] Miao W-C et al. 2024. Microdisplays: mini-LED, micro-OLED, and micro-LED. *Adv. Opt. Mater.* **12**, 2300112.
- [111] Park J et al. 2021. Electrically driven mid-submicrometre pixelation of InGaN micro-light-emitting diode displays for augmented-reality glasses. *Nat. Photon.* **15**, 449–455.
- [112] Song J H, Jiang J J, Li M H, Zhou Q X, Meng L Y, Sun K, Cui S Y, Wen K S, Liu S and Zhou S J. 2025. Recent advances in nitride-based Micro-LEDs for next-generation display. *Adv. Phys. Res.* **4**, 2400190.
- [113] Su Z H, Zhanghu M Y and Liu Z J. 2021. P-12.5: investigation on AR/VR displays based on novel Micro-LED technology. In *SID Symp. Digest of Technical Papers* vol 52, pp 609–612.
- [114] Day J, Li J, Lie D Y C, Bradford C, Lin J Y and Jiang H X. 2011. III-Nitride full-scale high-resolution microdisplays. *Appl. Phys. Lett.* **99**, 031116.
- [115] Bababekova Y, Rosenfield M, Hue J E and Huang R R. 2011. Font size and viewing distance of handheld smart phones. *Optom. Vis. Sci.* **88**, 795–797.
- [116] Hou J H, Nam Y, Peng W and Lee K M. 2012. Effects of screen size, viewing angle, and players' immersion tendencies on game experience. *Comput. Hum. Behav.* **28**, 617–623.
- [117] Jesty L C. 1958. The relation between picture size, viewing distance and picture quality. *Proc. IEEE B* **105**, 425–439.
- [118] Lombard M, Reich R D, Grabe M E, Bracken C C and Ditton T B. 2000. Presence and television: the role of screen size. *Hum. Commun. Res.* **26**, 75–98.
- [119] Long J, Cheung R, Duong S, Paynter R and Asper L. 2017. Viewing distance and eyestrain symptoms with prolonged viewing of smartphones. *Clin. Exp. Optom.* **100**, 133–137.
- [120] Shieh K-K and Lee D-S. 2007. Preferred viewing distance and screen angle of electronic paper displays. *Appl. Ergon.* **38**, 601–608.
- [121] Chen D B, Chen Y-C, Zeng G, Zhang D W and Lu H-L. 2023. Integration technology of micro-LED for next-generation display. *Research* **6**, 0047.
- [122] Bower C A, Menard E, Bonafede S, Hamer J W and Cok R S. 2011. Transfer-printed microscale integrated circuits for high performance display backplanes. *IEEE Trans. Compon. Packag. Manuf. Technol.* **1**, 1916–1922.
- [123] Carreira J F C, Guilhabert B J E, McKendry J J D, Xie E, Mathieson K, Watson I M, Gu E, Dawson M D and Henderson R K. 2018. Integration of Micro-LED array on CMOS by transfer printing. In *2018 IEEE Photonics Conference (IPC)* (IEEE, Reston, VA, USA) pp 1–2.
- [124] Cok R S et al. 2017. Inorganic light-emitting diode displays using micro-transfer printing. *J. Soc. Inform. Disp.* **25**, 589–609.
- [125] Kumar V, Behrman K and Kymissis I. 2024. Putting microLED technology on display. *Phys. Today* **77**, 30–36.
- [126] Meitl M et al. 2016. 55-1: invited Paper: passive matrix displays with transfer-printed microscale inorganic LEDs. In *SID Symp. Digest of Technical Papers* vol 47, pp 743–746.
- [127] Ahn J-H, Kim H-S, Lee K J, Jeon S, Kang S J, Sun Y G, Nuzzo R G and Rogers J A. 2006. Heterogeneous three-dimensional electronics by use of printed semiconductor nanomaterials. *Science* **314**, 1754–1757.
- [128] Bower C A, Meitl M and Kneeburg D. 2014. Micro-transfer-printing: heterogeneous integration of microscale semiconductor devices using elastomer stamps. In *SENSORS, 2014 IEEE* (IEEE, Valencia, Spain) pp 2111–2113.
- [129] Corbett B, Loi R, Zhou W D, Liu D and Ma Z Q. 2017. Transfer print techniques for heterogeneous integration of photonic components. *Prog. Quantum Electron.* **52**, 1–17.
- [130] Jeong J, Geum D-M and Kim S. 2022. Heterogeneous and monolithic 3D integration technology for mixed-signal ICs. *Electronics* **11**, 3013.
- [131] Lau J H. 2021. Chiplet heterogeneous integration. In *Semiconductor Advanced Packaging* (ed Lau J H) (Springer, Singapore) pp 413–439.
- [132] Liu Y, Huang Y and Duan X F. 2019. Van der Waals integration before and beyond two-dimensional materials. *Nature* **567**, 323–333.
- [133] Yoon J, Lee S-M, Kang D, Meitl M A, Bower C A and Rogers J A. 2015. Heterogeneously integrated optoelectronic devices enabled by micro-transfer printing. *Adv. Opt. Mater.* **3**, 1313–1335.
- [134] Kim D-H et al. 2011. Epidermal electronics. *Science* **333**, 838–843.
- [135] Lee S, Kang B, Keum H, Ahmed N, Rogers J A, Ferreira P M, Kim S and Min B. 2016. Heterogeneously assembled metamaterials and metadevices via 3D modular transfer printing. *Sci. Rep.* **6**, 27621.
- [136] Zhang H L et al. 2020. Heterogeneously integrated flexible microwave amplifiers on a cellulose nanofibril substrate. *Nat. Commun.* **11**, 3118.
- [137] Chun J, Lee K J, Leem Y-C, Kang W-M, Jeong T, Baek J H, Lee H J, Kim B-J and Park S-J. 2014. Vertically stacked color tunable light-emitting diodes fabricated using wafer bonding and transfer printing. *ACS Appl. Mater. Interfaces* **6**, 19482–19487.
- [138] Geum D-M, Kim S K, Kang C-M, Moon S-H, Kyhm J, Han J, Lee D-S and Kim S. 2019. Strategy toward the fabrication of ultrahigh-resolution micro-LED displays by bonding-interface-engineered vertical stacking and surface passivation. *Nanoscale* **11**, 23139–23148.
- [139] Kang C-M, Kong D-J, Shim J-P, Kim S, Choi S-B, Lee J-Y, Min J-H, Seo D-J, Choi S-Y and Lee D-S. 2017. Fabrication of a vertically-stacked passive-matrix micro-LED array structure for a dual color display. *Opt. Express* **25**, 2489–2495.
- [140] Kang C-M, Lee J-Y, Kong D-J, Shim J-P, Kim S, Mun S-H, Choi S-Y, Park M-D, Kim J and Lee D-S. 2018. Hybrid full-color inorganic light-emitting diodes integrated on a single wafer using selective area growth and adhesive bonding. *ACS Photonics* **5**, 4413–4422.
- [141] Li L Z et al. 2021. Transfer-printed, tandem microscale light-emitting diodes for full-color displays. *Proc. Natl Acad. Sci. USA* **118**, e2023436118.
- [142] Liao Z L and Mull D E. 1990. Wafer fusion: a novel technique for optoelectronic device fabrication and monolithic integration. *Appl. Phys. Lett.* **56**, 737–739.
- [143] Mun S-H, Kang C-M, Min J-H, Choi S-Y, Jeong W-L, Kim G-G, Lee J-S, Kim K-P, Ko H C and Lee D-S. 2021. Highly efficient full-color inorganic LEDs on a single wafer by using multiple adhesive bonding. *Adv. Mater. Interfaces* **8**, 2100300.
- [144] Ogihara M, Yokoyama S and Amemiya Y. 2019. Heterogeneous integration of GaN device layer by epitaxial film bonding. In *2019 2nd International Symposium on Devices, Circuits and Systems (ISDCS)* (IEEE, Higashi-Hiroshima, Japan) pp 1–3.
- [145] Shin J et al. 2023. Vertical full-colour micro-LEDs via 2D materials-based layer transfer. *Nature* **614**, 81–87.

- [146] Wong W S, Sands T, Cheung N W, Kneissl M, Bour D P, Mei P, Romano L T and Johnson N M. 2000. Inx Ga_{1-x}N light emitting diodes on Si substrates fabricated by Pd-In metal bonding and laser lift-off. *Appl. Phys. Lett.* **77**, 2822–2824.
- [147] El-Ghoroury H S, Yeh M, Chen J C, Li X and Chuang C-L. 2016. Growth of monolithic full-color GaN-based LED with intermediate carrier blocking layers. *AIP Adv.* **6**, 075316.
- [148] Even A, Laval G, Ledoux O, Ferret P, Sotta D, Guiot E, Levy F, Robin I C and Dussaigne A. 2017. Enhanced In incorporation in full InGa_N heterostructure grown on relaxed InGa_N pseudo-substrate. *Appl. Phys. Lett.* **110**, 262103.
- [149] He H H, Huang J P, Tao T, Zhi T, Zhang K X, Zhuang Z, Yan Y and Liu B. 2024. Monolithic integration of GaN-based transistors and Micro-LED. *Nanomaterials* **14**, 511.
- [150] Hwangbo S, Hu L, Hoang A T, Choi J Y and Ahn J-H. 2022. Wafer-scale monolithic integration of full-colour micro-LED display using MoS₂ transistor. *Nat. Nanotechnol.* **17**, 500–506.
- [151] Jung Hun C, Lee S J, Kwon K O, Choi J Y, Jung T, Han M and Han S J. 2020. A monolithically integrated micro-LED display based on GaN-on-silicon substrate. *Appl. Phys. Express.* **13**, 026501.
- [152] Lee Y-J, Yang Z-P, Chen P-G, Hsieh Y-A, Yao Y-C, Liao M-H, Lee M-H, Wang M-T and Hwang J-M. 2014. Monolithic integration of GaN-based light-emitting diodes and metal-oxide-semiconductor field-effect transistors. *Opt. Express* **22**, A1589–A1595.
- [153] Li Z, Waldron J, Detchprohm T, Wetzel C, Karlicek J R F and Chow T P. 2013. Monolithic integration of light-emitting diodes and power metal-oxide-semiconductor channel high-electron-mobility transistors for light-emitting power integrated circuits in GaN on sapphire substrate. *Appl. Phys. Lett.* **102**, 192107.
- [154] Liu C, Cai Y F, Zou X B and Lau K M. 2016. Low-leakage high-breakdown laterally integrated HEMT-LED via n-GaN electrode. *IEEE Photonics Technol. Lett.* **28**, 1130–1133.
- [155] Lu X, Liu C, Jiang H X, Zou X B, Zhang A P and Lau K M. 2016. Monolithic integration of enhancement-mode vertical driving transistor on a standard InGa_N/GaN light emitting diode structure. *Appl. Phys. Lett.* **109**, 053504.
- [156] Lu X, Yang S, Jiang H X and Wu J. 2019. Monolithic integration of GaN LEDs with vertical driving MOSFETs by selective area growth and band engineering of the p-AlGa_N electron blocking layer through TCAD simulation. *Semicond. Sci. Technol.* **34**, 064002.
- [157] Meng W Q et al. 2021. Three-dimensional monolithic micro-LED display driven by atomically thin transistor matrix. *Nat. Nanotechnol.* **16**, 1231–1236.
- [158] Tull B R, Basaran Z, Gidony D, Limanov A B, Im J S, Kyymissis I and Lee V W. 2015. 26.2: invited Paper: high brightness, emissive microdisplay by integration of III-V LEDs with thin film silicon transistors. *In SID Symp. Digest of Technical Papers* vol 46, pp 375–377.
- [159] Feng X, Meitl M A, Bowen A M, Huang Y G, Nuzzo R G and Rogers J A. 2007. Competing fracture in kinetically controlled transfer printing. *Langmuir.* **23**, 12555–12560.
- [160] Choi M, Jang B, Lee W, Lee S, Kim T W, Lee H-J, Kim J-H and Ahn J-H. 2017. Stretchable active matrix inorganic light-emitting diode display enabled by overlay-aligned roll-transfer printing. *Adv. Funct. Mater.* **27**, 1606005.
- [161] Wang C J et al. 2020. Programmable and scalable transfer printing with high reliability and efficiency for flexible inorganic electronics. *Sci. Adv.* **6**, eabb2393.
- [162] Park S H et al. 2023. Universal selective transfer printing via micro-vacuum force. *Nat. Commun.* **14**, 7744.
- [163] Chang W et al. 2023. Concurrent self-assembly of RGB microLEDs for next-generation displays. *Nature* **617**, 287–291.
- [164] Ahn J H, Meitl M A, Baca A J, Khang D Y, Kim H S and Rogers J A. 2008. *Solution Processing of Inorganic Materials*. (John Wiley & Sons, Inc) pp 407–447.
- [165] Carlson A. 2012. *Advanced Stamp Geometries and Adhesiveless Transfer Printing Modalities for Use in Deterministic Materials Assembly*. (University of Illinois at Urbana-Champaign, Urbana, USA).
- [166] Carlson A, Bowen A M, Huang Y G, Nuzzo R G and Rogers J A. 2012. Transfer printing techniques for materials assembly and micro/nanodevice fabrication. *Adv. Mater.* **24**, 5284–5318.
- [167] Linghu C H, Zhang S, Wang C J and Song J Z. 2018. Transfer printing techniques for flexible and stretchable inorganic electronics. *npj Flex. Electron.* **2**, 26.
- [168] Zhou H L, Qin W Y, Yu Q M, Cheng H Y, Yu X D and Wu H P. 2019. Transfer printing and its applications in flexible electronic devices. *Nanomaterials* **9**, 283.
- [169] Choi M K et al. 2015. Wearable red–green–blue quantum dot light-emitting diode array using high-resolution intaglio transfer printing. *Nat. Commun.* **6**, 7149.
- [170] Chung H-J, Kim T-I, Kim H-S, Wells S A, Jo S, Ahmed N, Jung Y H, Won S M, Bower C A and Rogers J A. 2011. Fabrication of releasable single-crystal silicon–metal oxide field-effect devices and their deterministic assembly on foreign substrates. *Adv. Funct. Mater.* **21**, 3029–3036.
- [171] Hwang S-W et al. 2012. A physically transient form of silicon electronics. *Science.* **337**, 1640–1644.
- [172] Jeong J W et al. 2014. High-resolution nanotransfer printing applicable to diverse surfaces via interface-targeted adhesion switching. *Nat. Commun.* **5**, 5387.
- [173] Jung Y H et al. 2015. High-performance green flexible electronics based on biodegradable cellulose nanofibril paper. *Nat. Commun.* **6**, 7170.
- [174] Jung Y H, Zhang H L, Lee I-K, Shin J H, Kim T-I and Ma Z Q. 2019. Releasable high-performance GaAs schottky diodes for gigahertz operation of flexible bridge rectifier. *Adv. Electron. Mater.* **5**, 1800772.
- [175] Kim B H et al. 2016. Multilayer transfer printing for pixelated, multicolor quantum dot light-emitting diodes. *ACS Nano* **10**, 4920–4925.
- [176] Kim D-H, Ahn J-H, Choi W M, Kim H-S, Kim T-H, Song J Z, Huang Y Y, Liu Z J, Lu C and Rogers J A. 2008. Stretchable and foldable silicon integrated circuits. *Science* **320**, 507–511.
- [177] Kim D-H, Kim Y-S, Wu J, Liu Z J, Song J Z, Kim H-S, Huang Y Y, Hwang K-C and Rogers J A. 2009. Ultrathin silicon circuits with strain-isolation layers and mesh layouts for high-performance electronics on fabric, vinyl, leather, and paper. *Adv. Mater.* **21**, 3703–3707.
- [178] Kim D-H et al. 2010. Dissolvable films of silk fibroin for ultrathin conformal bio-integrated electronics. *Nat. Mater.* **9**, 511–517.
- [179] Kim T-H et al. 2011. Full-colour quantum dot displays fabricated by transfer printing. *Nat. Photon.* **5**, 176–182.
- [180] Kim T-I, Hwan Jung Y, Chung H-J, Jun Yu K, Ahmed N, Corcoran C J, Suk Park J, Hun Jin S and Rogers J A. 2013. Deterministic assembly of releasable single crystal silicon-metal oxide field-effect devices formed from bulk wafers. *Appl. Phys. Lett.* **102**, 182104.
- [181] Kim T-I et al. 2014. Thin film receiver materials for deterministic assembly by transfer printing. *Chem. Mater.* **26**, 3502–3507.
- [182] Kim T-I et al. 2013. Injectable, cellular-scale optoelectronics with applications for wireless optogenetics. *Science* **340**, 211–216.

- [183] Kim-Lee H-J, Carlson A, Grierson D S, Rogers J A and Turner K T. 2014. Interface mechanics of adhesiveless microtransfer printing processes. *J. Appl. Phys.* **115**, 143513.
- [184] Liu C et al. 2017. Bendable MOS capacitors formed with printed $\text{In}_{0.2}\text{Ga}_{0.8}\text{As}/\text{GaAs}/\text{In}_{0.2}\text{Ga}_{0.8}\text{As}$ trilayer nanomembrane on plastic substrates. *Appl. Phys. Lett.* **110**, 133505.
- [185] McCall J G, Kim T-I, Shin G, Huang X, Jung Y H, Al-Hasani R, Omenetto F G, Bruchas M R and Rogers J A. 2013. Fabrication and application of flexible, multimodal light-emitting devices for wireless optogenetics. *Nat. Protoc.* **8**, 2413–2428.
- [186] Yeo W-H et al. 2013. Multifunctional epidermal electronics printed directly onto the skin. *Adv. Mater.* **25**, 2773–2778.
- [187] Yoon J et al. 2008. Ultrathin silicon solar microcells for semitransparent, mechanically flexible and microconcentrator module designs. *Nat. Mater.* **7**, 907–915.
- [188] Kim T-H, Carlson A, Ahn J-H, Won S M, Wang S D, Huang Y G and Rogers J A. 2009. Kinetically controlled, adhesiveless transfer printing using microstructured stamps. *Appl. Phys. Lett.* **94**, 113502.
- [189] Meitl M A, Zhu Z-T, Kumar V, Lee K J, Feng X, Huang Y Y, Adesida I, Nuzzo R G and Rogers J A. 2006. Transfer printing by kinetic control of adhesion to an elastomeric stamp. *Nat. Mater.* **5**, 33–38.
- [190] Ahmed N, Dagdeviren C, Rogers J A and Ferreira P M. 2015. Active polymeric composite membranes for localized actuation and sensing in microtransfer printing. *J. Microelectromech. Syst.* **24**, 1016–1028.
- [191] Bower C A et al. 2017. Emissive displays with transfer-printed assemblies of $8\mu\text{m} \times 15\mu\text{m}$ inorganic light-emitting diodes. *Photon. Res.* **5**, A23–A29.
- [192] Gomez D et al. 2016. Process capability and elastomer stamp lifetime in micro transfer printing. In *2016 IEEE 66th Electronic Components and Technology Conference (ECTC)* (IEEE, Las Vegas, NV, USA) pp 680–687.
- [193] Kim R-H, Kim S, Song Y M, Jeong H, Kim T-I, Lee J, Li X L, Choquette K D and Rogers J A. 2012. Flexible vertical light emitting diodes. *Small* **8**, 3123–3128.
- [194] Loo Y-L, Hsu J W P, Willett R L, Baldwin K W, West K W and Rogers J A. 2002. High-resolution transfer printing on GaAs surfaces using alkane dithiol monolayers. *J. Vac. Sci. Technol. B* **20**, 2853–2856.
- [195] Lueck M, Huffman A, Hines P, Lannon J, Bonafede S, Trindade A J and Bower C A. 2016. Fan-out packaging of microdevices assembled using micro-transfer-printing. In *2016 IEEE 66th Electronic Components and Technology Conference (ECTC)* (IEEE, Las Vegas, NV, USA) pp 37–42.
- [196] Park S-I et al. 2009. Printed assemblies of inorganic light-emitting diodes for deformable and semitransparent displays. *Science* **325**, 977–981.
- [197] Kim S et al. 2010. Microstructured elastomeric surfaces with reversible adhesion and examples of their use in deterministic assembly by transfer printing. *Proc. Natl Acad. Sci. USA*. **107**, 17095–17100.
- [198] Kim S, Carlson A, Cheng H Y, Lee S, Park J-K, Huang Y G and Rogers J A. 2012. Enhanced adhesion with pedestal-shaped elastomeric stamps for transfer printing. *Appl. Phys. Lett.* **100**, 171909.
- [199] Carlson A, Wang S D, Elvikis P, Ferreira P M, Huang Y G and Rogers J A. 2012. Active, programmable elastomeric surfaces with tunable adhesion for deterministic assembly by transfer printing. *Adv. Funct. Mater.* **22**, 4476–4484.
- [200] Carlson A et al. 2011. Shear-enhanced adhesiveless transfer printing for use in deterministic materials assembly. *Appl. Phys. Lett.* **98**, 264104.
- [201] Eisenhaure J D, Rhee S I, Al-Okaily A M, Carlson A, Ferreira P M and Kim S. 2014. The use of shape memory polymers for microassembly by transfer printing. *J. Microelectromech. Syst.* **23**, 1012–1014.
- [202] Eisenhaure J D, Xie T, Varghese S and Kim S. 2013. Microstructured shape memory polymer surfaces with reversible dry adhesion. *ACS Appl. Mater. Interfaces* **5**, 7714–7717.
- [203] Lee H, Um D-S, Lee Y, Lim S, Kim H-J and Ko H. 2016. Octopus-inspired smart adhesive pads for transfer printing of semiconducting nanomembranes. *Adv. Mater.* **28**, 7457–7465.
- [204] Mengüç Y, Yang S Y, Kim S, Rogers J A and Sitti M. 2012. Gecko-inspired controllable adhesive structures applied to micromanipulation. *Adv. Funct. Mater.* **22**, 1246–1254.
- [205] Yoo B, Cho S, Seo S and Lee J. 2014. Elastomeric angled microflaps with reversible adhesion for transfer-printing semiconductor membranes onto dry surfaces. *ACS Appl. Mater. Interfaces* **6**, 19247–19253.
- [206] Yang Y M, Hwang Y, Cho H A, Song J-H, Park S-J, Rogers J A and Ko H C. 2011. Arrays of silicon micro/nanostructures formed in suspended configurations for deterministic assembly using flat and roller-type stamps. *Small* **7**, 484–491.
- [207] Yang S Y et al. 2012. Elastomer surfaces with directionally dependent adhesion strength and their use in transfer printing with continuous roll-to-roll applications. *Adv. Mater.* **24**, 2117–2122.
- [208] Yoon S-U, Hwangbo Y, Jang B, Kim H-D and Kim J-H. 2024. Transfer of Micro-LEDs with roll-based direct overlay alignment for manufacturing transparent displays. *Adv. Electron. Mater.* **10**, 2400236.
- [209] Hu L, Choi J, Hwangbo S, Kwon D-H, Jang B, Ji S, Kim J-H, Han S-K and Ahn J-H. 2022. Flexible micro-LED display and its application in Gbps multi-channel visible light communication. *npj Flex. Electron.* **6**, 100.
- [210] Jang B, Won S, Kim J, Kim J, Oh M, Lee H-J and Kim J-H. 2022. Auxetic meta-display: stretchable display without image distortion. *Adv. Funct. Mater.* **32**, 2113299.
- [211] Kim D H, Yoo H G, Hong S M, Jang B, Park D Y, Joe D J, Kim J-H and Lee K J. 2016. Simultaneous roll transfer and interconnection of flexible silicon NAND flash memory. *Adv. Mater.* **28**, 8371–8378.
- [212] Park S-C, Fang J, Biswas S, Mozafari M, Stauden T and Jacobs H O. 2015. Approaching roll-to-roll fluidic self-assembly: relevant parameters, machine design, and applications. *J. Microelectromech. Syst.* **24**, 1928–1937.
- [213] Sharma B K, Jang B, Lee J E, Bae S-H, Kim T W, Lee H-J, Kim J-H and Ahn J-H. 2013. Load-controlled roll transfer of oxide transistors for stretchable electronics. *Adv. Funct. Mater.* **23**, 2024–2032.
- [214] Bibl A and Golda D. 2014. Compliant micro device transfer head with integrated electrode leads. *U.S. Patent*. No. 8,945,968B2.
- [215] Bibl A and Golda D. 2016. Mass transfer system. *U.S. Patent*. No. 9,484,237-B2.
- [216] Bibl A, Higginson J A, Hu H H and Law H F S. 2017. Method of transferring and bonding an array of micro devices. *U.S. Patent*. No. 9,773,750-B2.
- [217] Bibl A, Higginson J A, Law H F S and Hu H H. 2012. Method of transferring a micro device. *U.S. Patent*. No. 8,338,601-B1.
- [218] Bibl A, Higginson J A, Law H F S and Hu H H. 2013. Micro device transfer head heater assembly and method of transferring a micro device. *U.S. Patent*. No. 8,349,116-B1.
- [219] Golda D, Higginson J A, Bibl A, Parks P A and Bathurst S P. 2015. Micro pick up array mount with integrated displacement sensor. *U.S. Patent*. No. 9,095,980-B2.

- [220] Hu H H and Bibl A 2015. Stabilization structure including sacrificial release layer and staging cavity. *U.S. Patent*. No. 9166114-B2.
- [221] Wu M H, Fang Y H and Chao C H 2017. Electric-programmable magnetic module and picking-up and placement process for electronic devices. *U.S. Patent*. No. 9607907-B2.
- [222] Bian J, Zhou L B Y, Wan X D, Zhu C, Yang B and Huang Y A. 2019. Laser transfer, printing, and assembly techniques for flexible electronics. *Adv. Electron. Mater.* **5**, 1800900.
- [223] Saeidpourazar R, Sangid M D, Rogers J A and Ferreira P M. 2012. A prototype printer for laser driven micro-transfer printing. *J. Manuf. Process.* **14**, 416–424.
- [224] Al-okaily A M and Ferreira P M. 2015. Multi-physics modeling for laser micro-transfer printing delamination. *J. Manuf. Process.* **20**, 414–424.
- [225] Luo H Y, Wang C J, Zhang S and Song J Z 2024. Laser-driven transfer printing techniques for micro-LED display. In *Transfer Printing Technologies and Applications* (eds Cao C H, Sun Y) (Elsevier, Amsterdam) pp 325–351.
- [226] Luo H Y, Wang S H, Wang C J, Linghu C H and Song J Z. 2021. Thermal controlled tunable adhesive for deterministic assembly by transfer printing. *Adv. Funct. Mater.* **31**, 2010297.
- [227] Chang Y-K and Hong F C-N. 2009. The fabrication of ZnO nanowire field-effect transistors by roll-transfer printing. *Nanotechnology* **20**, 195302.
- [228] Chen F R, Gai M X, Sun N N, Xu Z Y, Liu L, Yu H Y, Bian J and Huang Y A. 2023. Laser-driven hierarchical “gas-needles” for programmable and high-precision proximity transfer printing of microchips. *Sci. Adv.* **9**, eadk0244.
- [229] Eisenhaure J and Kim S. 2016. Laser-driven shape memory effect for transfer printing combining parallelism with individual object control. *Adv. Mater. Technol.* **1**, 1600098.
- [230] Gao Y Y, Li Y H, Li R and Song J Z. 2017. An accurate thermomechanical model for laser-driven microtransfer printing. *J. Appl. Mech.* **84**, 064501.
- [231] Huang Y, Zheng N, Cheng Z Q, Chen Y, Lu B W, Xie T and Feng X. 2016. Direct laser writing-based programmable transfer printing via bioinspired shape memory reversible adhesive. *ACS Appl. Mater. Interfaces* **8**, 35628–35633.
- [232] Lang T F et al. 2025. Thin-film assisted laser transfer and bonding (TFA-LTAB) for the fabrication of Micro-LED displays. *Adv. Electron. Mater.* **11**, 2400380.
- [233] Li C L, Zhang S, Jiang J, Wang S H, He S C and Song J Z. 2024. Laser-induced adhesives with excellent adhesion enhancement and reduction capabilities for transfer printing of microchips. *Sci. Adv.* **10**, eads9226.
- [234] Lampen Shaun I R, Khurana R N, Noronha G, Brown D M and Wykoff C C. 2018. 52-4: laser-enabled extremely-high rate technology for μ LED assembly. In *SID Symp. Digest of Technical Papers* vol 49, pp 692–695.
- [235] Saeidpourazar R, Li R, Li Y H, Sangid M D, Lu C F, Huang Y G, Rogers J A and Ferreira P M. 2012. Laser-driven micro transfer placement of prefabricated microstructures. *J. Microelectromech. Syst.* **21**, 1049–1058.
- [236] Valentine A D, Busbee T A, Boley J W, Raney J R, Chortos A, Kotikian A, Berrigan J D, Durstock M F and Lewis J A. 2017. Hybrid 3D printing of soft electronics. *Adv. Mater.* **29**, 1703817.
- [237] Jacobs H O, Tao A R, Schwartz A, Gracias D H and Whitesides G M. 2002. Fabrication of a cylindrical display by patterned assembly. *Science* **296**, 323–325.
- [238] Yeh H-J-J and Smith J S. 1994. Fluidic self-assembly for the integration of GaAs light-emitting diodes on Si substrates. *IEEE Photonics Technol. Lett.* **6**, 706–708.
- [239] Hwang J et al. 2023. Wafer-scale alignment and integration of micro-light-emitting diodes using engineered van der Waals forces. *Nat. Electron.* **6**, 216–224.
- [240] Lee D et al. 2023. Fluidic self-assembly for MicroLED displays by controlled viscosity. *Nature* **619**, 755–760.
- [241] Lee J S, Kang S J, Shin J H, Shin Y J, Lee B, Koo J-M and Kim T-I. 2020. Nanoscale-dewetting-based direct interconnection of microelectronics for a deterministic assembly of transfer printing. *Adv. Mater.* **32**, 1908422.
- [242] Lee S G et al. 2025. Mechanically stable, and reversible integration of microchips on textile: liquid metal-based anisotropic conductive adhesive. *npj Flex. Electron.* **9**, 72.
- [243] Chen E G, Fan Z G, Zhang K X, Huang C L, Xu S, Ye Y, Sun J, Yan Q and Guo T L. 2024. Broadband beam collimation metasurface for full-color micro-LED displays. *Opt. Express* **32**, 10252–10264.
- [244] Chen E G, Yao Z M, Fan Z G, Lai W Z, Liang T, Yan Q, Guo T L and Chen C P. 2024. Collimated LED array with mushroom-cap encapsulation for near-eye display projection engine. *IEEE J. Sel.* **30**, 2000410.
- [245] Jin H J et al. 2025. Micro-LED retinal projection for augmented reality near-eye displays. *Laser Photonics Rev.* **19**, 2402083.
- [246] Liu Y Z, Xia T W, Du A C, Liang T, Fan Z G, Chen E G, Sun J, Yan Q and Guo T L. 2023. Omnidirectional color shift suppression of full-color micro-LED displays with enhanced light extraction efficiency. *Opt. Lett.* **48**, 1650–1653.
- [247] Virey E H and Bouhamri Z. 2021. 59-1: invited Paper: from lab to fab: challenges and requirements for high-volume MicroLED manufacturing equipment. In *SID Symp. Digest of Technical Papers* vol 52, pp 826–829.
- [248] Choi K S, Joo J, Eom Y S, Choi G M, Jang K S, Lee C, Moon S H, Yun H G, Choi J H and Choi J W 2021. Simultaneous transfer and bonding (SITRAB) process for Micro-LEDs using laser-assisted bonding with compression (LABC) process and SITRAB adhesive. In *2021 IEEE 71st Electronic Components and Technology Conference (ECTC)* (IEEE, San Diego, CA, USA) pp 1607–1613.